

Supplementary Information for Proteus: Adaptive Scale-Space Analysis for Learning Non-Linear Fuzzy Manifold Memberships

S0. Notation and Pipeline Contract

S0.1 Global Notation

The table below summarizes the recurring symbols used across the SI and points to the sections where their canonical definitions are developed.

Symbol	Meaning	Owner
$\mathbf{x}$	data sample	S0
C, C_ℓ, C'	region, query-path region, child region	S0.2
τ, τ_C^*	candidate and selected variance cap	S2.4–S2.5.1
$u_{\text{global}}, u_{\text{local}}$	global and local tissue floors	S0.6
Ψ, ψ_i	local noise tensor and node-level floor proxy	S2.5.2–S2.5.3
$G_C = (\mathcal{V}_C, \mathcal{E}_C)$	Stage 1 scaffold graph	S2
$\mathcal{K}_C, \mathcal{S}_C$	Stage 2 complex and simplex set	S4
$S, S _d$	simplex and intrinsic d -volume	S4,S6
$h_i, n_{i \rightarrow j}$	hit mass and directed transition count	S2,S3
$\mathbf{m}_i, \mathbf{s}_i, \sigma_i^2$	residual moments and local variance	S2
m_S	simplex mass	S10
$\hat{p}_f, p_f$	empirical and reconstructed face pressure	S6
T_C	optional region warp	S5,S7
μ_C, Σ_C	Gaussian mode and covariance	S7
$\mu_C, \mu_C^{\text{PL}}, \mu(\mathbf{x})$	Gaussian, PL, and path memberships	S7
$\Phi_C(\tau), R_i(\tau)$	cluster and node scale responses	S2.5
$\alpha_{0,i}$	BDeu concentration	S2.7
$F_{\text{DM}}, \tau_{\text{BF}}$	evidence score and Bayes-factor margin	S3.4
W, T_{hyst}, A_T	gate window, hysteresis, jump budget	S3.6,S12
J_i	junction score	S8.4
$A_S, \tilde{\rho}_S, \bar{w}_S$	divergence stencil and PL density profile	S6

S0.2 Per-Region Convention

Every scaffold, complex, count table, mass field, pressure field, and Gaussian summary is owned by a recursion-tree region C . Unless a section explicitly discusses cross-level or cross-region behavior, it is written in the local convention: fix a region C and omit the superscript (C) from all objects. When comparing parent/child regions, sibling regions, or multiple query levels, superscripts are restored.

S0.2.1 Superscript Discipline

The superscript convention distinguishes single-region statements from statements over the recursion tree.

- Unsuperscripted symbols such as m_S , p_f , $\mathcal{K}$, and τ^* refer to the single region C fixed by the surrounding section.
- Superscripted symbols such as $m_S^{(C)}$, $p_f^{(C)}$, $\mathcal{K}_C$, and τ_C^* appear when a statement ranges over multiple regions in the recursion tree.
- C_ℓ denotes the ℓ -th region on a query path, while C' denotes a child region produced by recursive decomposition.

This convention distinguishes whether a mass, pressure, or membership score belongs to a single complex or to the multiscale tree.

S0.3 The Three Field Types

Proteus routes information through three fields with different schedules and meanings.

1. *Transition counts* $\{n_{i \rightarrow j}\}$ are empirical observations. They are written online by BMU/neighbor routing and read by Affinity Propagation, pruning diagnostics, and the Dirichlet–multinomial evidence gate.
2. *Simplex masses* $\{m_S\}$ are latent generative variables on an accepted complex. They summarize how probability mass is distributed over cells after topology selection and settling. The runtime evidence gate does not require an iterative MAP estimate of m ; it scores transition counts directly under a local Dirichlet–multinomial marginal.
3. *Face pressures* $\{p_f\}$ are interpolation variables on facets. Empirical tallies $\{\hat{p}_f\}$ are accumulated online, while the conservative field $\{p_f\}$ is reconstructed after topology selection. Pressures are used for density interpolation and diagnostics, not as a second likelihood term for the same edit decision.

The separation is architectural: counts decide structure; masses and pressures make the accepted structure queryable.

S0.4 Propose–Test–Settle Contract

All structural changes follow the same lifecycle.

1. *Propose*: a local diagnostic identifies a candidate edit. Examples include a variance-cap split, a weak-link prune, a merge, a torsion-aligned split, or a local warp attachment.
2. *Test*: the affected region is rerouted under the candidate topology and scored by the local evidence criterion F_{DM} . Evidence, not the diagnostic alone, decides whether the edit is accepted.
3. *Settle*: accepted edits trigger a local re-equilibration window so subsequent proposals are scored against stabilized counts and moments.

This contract is the shared assumption behind the statistical gates (S3), the Stage 2 construction rules (S4), torsion/warp proposals (S5), and the dynamic-convergence arguments (S12).

S0.5 Canonical Definition Sites

Several objects appear in multiple parts of Proteus. The following list identifies where their canonical definitions are developed.

- $\Phi_C(\tau)$ and $R_i(\tau)$: S2.5. R_i is a scale diagnostic, not a within-cluster density signal.

- $\alpha_{0,i}$: S2.7, derived from the BDeu prior and reused by AP preferences and the evidence gate.
- F_{DM} : S3.4, with consistency in S3.5.
- The flag-complex invariant and transition-count policies: S4.1–S4.4.
- Torsion ratio R_S , edge-ratio fallback, and warp policies: S5.
- Face pressures p_f and simplex-local density: S6.
- Gaussian summaries $(\boldsymbol{\mu}_C, \Sigma_C)$, closed-form membership $\mu_C(\mathbf{x})$, PL membership μ_C^{PL} , and trajectory $\boldsymbol{\mu}(\mathbf{x})$: S7.
- Generative mass field m , geometry-derived router $q(j \mid i; m)$, and identifiability: S10.
- Lyapunov function and dynamic convergence lemmas: S12.

Later sections may restate these objects locally for readability, but the sections listed above contain the full definitions and operational conventions.

S0.6 Noise Floor Decomposition

Stage 1 operates in a noise-positive domain. The latent density is modeled as

$$\rho(\mathbf{x}) = u(\mathbf{x}) + \sum_{i=1}^m f_i(\mathbf{x}), \quad u(\mathbf{x}) \geq u_{\min} > 0,$$

where the strictly positive tissue term $u(\mathbf{x})$ is the contrast reference against which all features are judged. It is useful to split this floor into

$$u(\mathbf{x}) = u_{\text{global}} + u_{\text{local}}(\mathbf{x}), \quad u_{\text{global}} > 0, \quad u_{\text{local}}(\mathbf{x}) \geq 0.$$

The global floor is the ambient-space minimum induced by irreducible measurement stochasticity; it determines whether any structure is detectable at all. The local floor is the excess tissue specific to a region or subregion: unresolved sub-structure, local contamination, or transverse blur from nearby features. Once recursion isolates a region, u_{local} becomes the relevant detectability reference.

Operationally, detectability and resolvability are always interpreted relative to the floor appropriate to the current recursion context. The root region sees the global floor; a recursion child sees the local floor within its parent interior. The purpose of Stage 1 local-noise tracking is not to estimate an arbitrary covariance model everywhere, but to maintain a conservative, confidence-gated local floor summary sufficient for scale-selection, cluster detection, and recursion decisions.

S1. Preprocessing and Optional Dimensionality Reduction

S1.1 Randomized SVD Preprocessing

For very high-dimensional inputs, Proteus may begin with an optional randomized SVD projection. The default protocol samples up to 20,000 rows (or all rows if the data are small), runs a randomized SVD with sketch width $k_{\text{target}} \approx 256$ and oversampling 20, and selects the retained rank by cumulative explained energy and an intrinsic-dimension floor:

$$r = \max(r_{0.999}, 2 \lceil D_{\text{LB}} \rceil),$$

where $r_{0.999}$ is the smallest rank whose cumulative explained variance exceeds 99.9% and D_{LB} is a Levina–Bickel intrinsic-dimension estimate. The projection is applied as $Z = XP$ with

$P = V_{1:r}^\top$. If the implied reduction is below 5% or near-neighbor recall on a held-out subsample drops materially, skip the projection. Optionally append 32 Johnson–Lindenstrauss vectors to tighten distance preservation. The projection is a computational preconditioner only: all later scale and topology decisions are still made by the Proteus scaffold.

S1.2 Optional Marginal Gaussianization

For highly skewed or heavy-tailed coordinates, an optional marginal Gaussianization can improve early neighbor search and moment estimates. The transform is applied in the current working representation: after the S1.1 projection if one is used, or on the original feature coordinates if projection is skipped. A coordinate is eligible only if it has enough observed support for a stable empirical CDF and behaves as a continuous scalar rather than a categorical or nearly binary variable. In first implementations, require at least $\max(1000, 10d)$ non-missing observations and at least 50 distinct values; coordinates that fail these checks are left unchanged.

Among eligible coordinates, transform only those whose marginal distribution is meaningfully non-Gaussian, e.g. large absolute skew, heavy tails, or an Anderson–Darling / Kolmogorov–Smirnov rejection of normality on a held-out subsample. For each selected coordinate u_j , estimate an empirical CDF $\hat{F}_j$ on the training split and transform

$$z_j = \sqrt{2} \operatorname{erf}^{-1} \left(2\hat{F}_j(u_j) - 1 \right).$$

The fitted CDF is then frozen and applied to validation/test samples by interpolation with clipped tail probabilities, e.g. $\hat{F}_j \in [\epsilon, 1 - \epsilon]$ with $\epsilon = 1/(2n_j)$. This makes each selected marginal approximately standard normal on the training sample, but it is still only a marginal transform; it does not claim to linearize the joint manifold.

The transform is kept only if it preserves the geometry Proteus needs. On a held-out subsample, compare nearest-neighbor recall and rank correlation of local distances before and after the transform; reject the transform if recall drops materially (default: more than 5% absolute at the Stage 1 k) or if local distance rank correlation falls below 0.95. Also run lightweight linearity diagnostics on the transformed space (graph-Laplacian curvature energy and distance correlation are good defaults) and require them to improve or remain statistically indistinguishable. If the linearity test passes at $p \geq 0.01$, the expensive global warp fitting can be skipped at this recursion level and the algorithm can proceed directly to the geometric audit. If the transform worsens neighbor preservation, destabilizes the scale-response peaks, or increases Stage 1 stabilization time, it is rolled back for that recursion level.

S1.3 Distance Preservation and Failure Modes

Preprocessing is skipped if nearest-neighbor preservation deteriorates, if the retained dimension is not materially smaller, or if downstream scale-response diagnostics become less stable. Implementations should log local neighbor recall before/after projection, explained variance, retained rank, and any change in Stage 1 stabilization time. The default posture is conservative: preprocessing is a speed optimization, not part of the statistical guarantee.

S1.4 Per-Recursion Intrinsic-Dimension Projection

At every parent-to-child recursion boundary, Proteus re-expresses each child region in a local PCA frame before the child’s next Stage 1 scale search. This step is mandatory: optional marginal Gaussianization and non-linear warps are both fitted and evaluated after this transfer, in the child PCA frame. The PCA therefore fixes the working coordinates for the recursion level; later transforms add Gaussianization or local non-linear capacity without replacing the inherited intrinsic-dimension projection. The PCA is fitted in the active parent frame, meaning after any accepted parent-level projection or warp has already been applied.

Let C' be a child region produced by AP clustering in parent region C , and let $\mathcal{V}_{C'}$ be the parent scaffold nodes assigned to that child. Define the child intrinsic-dimension summary by the cluster median

$$\widehat{d}_{C'} = \text{median}\{d_{\text{final},i} : i \in \mathcal{V}_{C'}\}.$$

The target rank is

$$r_{\text{ID}}(C') = \max\left(r_{\min}, 2\lceil\widehat{d}_{C'}\rceil\right),$$

with default $r_{\min} = 3$ for non-degenerate local neighborhoods. Run a PCA or randomized SVD on the centered routed child samples in the active parent frame and compute the usual explained-energy rank $r_{0.999}(C')$. The retained child working rank is

$$r_{C'} = \min\left(D_C, \max\{r_{\text{ID}}(C'), r_{0.999}(C')\}\right),$$

where D_C is the parent working dimension. Child samples and inherited node positions are projected into the first $r_{C'}$ principal directions before recursion begins, and the child's D_{subspace} is set to $r_{C'}$.

The distance-preservation audit of S1.3 is still applied locally. If truncating to $r_{C'}$ materially degrades nearest-neighbor recall or local distance ranks, keep the full child PCA basis in the parent dimension and log the failed truncation; the mandatory operation is the child PCA reparameterization, while rank truncation remains guarded by neighbor preservation. Child-level marginal Gaussianization and all subsequent region or patch warp proposals are then fitted in this PCA frame. This makes those transforms cheaper because their input dimension is $r_{C'}$ rather than the parent dimension, while the explained-energy and neighbor-preservation guards ensure that no accepted child-level information is discarded by the rank choice. Along a root-to-leaf path, projections compose with the parent representation, so each child works inside the span made available by its ancestors and no downstream level invents directions outside the accepted parent frame.

S2. Stage 1 Derivations

S2.1 Geometric Grid Spacing

Lindeberg-normalized scale-space responses obey $\Delta_{\text{FWHM}}(\log \sigma) \approx 0.586$: peaks separated by less than this in log-bandwidth are not reliably resolvable. Choose adjacent grid scales separated by half the FWHM,

$$\log \sigma_{j+1} - \log \sigma_j = \Delta \approx \frac{1}{2} \Delta_{\text{FWHM}} \implies \frac{\sigma_{j+1}}{\sigma_j} = e^{\Delta} \approx \sqrt{2}.$$

Since $\tau = \sigma^2$, the corresponding grid ratio in the variance cap is $\tau_{j+1}/\tau_j \approx 1/2$. In practice we use the in- σ ratio $r = 1/\sqrt{2} \approx 0.71$ (about 7–9 points per decade) and map consistently into τ . This is the resolution criterion behind the default $r = 1/\sqrt{2}$.

The geometric grid is the canonical bookkeeping lattice, not the full search policy. In the presence of a locally varying floor, Stage 1 should treat the grid as the set of admissible *candidate* scales and adapt traversal to the current local floor belief. Coarser scales remain attractive while a region is still merged into composite structure; finer scales are favored only after the current scale is both stabilized and locally floor-feasible. In practice, the adaptive policy is driven by the local floor summaries of S2.5.2–S2.5.3: the geometric grid stays fixed, but the controller's confidence in each step becomes floor-aware rather than globally homoscedastic.

S2.2 EWMA Weight from Neighborhood Size

The EWMA recursion $m_{t+1} = (1 - \alpha)m_t + \alpha x_{t+1}$ has half-life $T_{1/2} = \ln(0.5)/\ln(1 - \alpha) \approx \ln 2/\alpha$ for small α . Setting $T_{1/2} = k$ (one effective neighborhood of updates) yields

$$\alpha = \frac{\ln 2}{k}.$$

This ties estimator memory to the effective neighborhood support: changing k scales the half-life by the same factor without altering the bias/variance interpretation of the moments. Equivalently, with effective neighborhood mass H_{eff} , the standard half-window rule $\alpha \approx 2/(H_{\text{eff}} + 1)$ is recovered when $H_{\text{eff}} \approx 2k/\ln 2 - 1$.

S2.3 Dual-Rate Motion from Variance Correction

Stage 1 separates fast statistical tracking from slower geometric movement. For sample $\mathbf{x}$ and node i , define residual $\mathbf{e}_i = \mathbf{x} - \mathbf{w}_i$. Moment updates track $\mathbb{E}[\mathbf{e}_i]$ and $\mathbb{E}[\mathbf{e}_i \mathbf{e}_i^\top]$ immediately. Geometric motion is driven by the EWMA residual mean $\mathbf{m}_i$ rather than by raw per-sample residuals: $\mathbf{m}_i$ estimates persistent local bias, while the deferred nudge variable $\mathbf{a}_i$ accumulates only scale-resolved displacement before committing it to the node position. Only when $\|\mathbf{a}_i\|$ exceeds a displacement threshold is $\mathbf{w}_i$ updated and the ANN index refreshed. This dual-rate design keeps neighbor search stable and allows convergence arguments to reason about virtual positions $\mathbf{y}_i = \mathbf{w}_i + \mathbf{a}_i$ between realized moves.

Node state. Each node stores hit count h_i , residual mean $\mathbf{m}_i$, residual second moment $\mathbf{s}_i$, derived variance σ_i^2 , an Oja direction $\mathbf{u}_i$ for dominant local variance, directed transition counts $n_{i \rightarrow j}$, and deferred nudge $\mathbf{a}_i$. A variance-cap violation $\sigma_i^2 > \tau_{\text{local},i}$ proposes additional local capacity, typically by splitting along $\mathbf{u}_i$ after evidence and budget checks. Low-hit, low-variance nodes or weak bilateral links become prune candidates. The variance cap is therefore both a geometric resolution target and a proposal generator; the evidence gate decides whether the proposed topology change is retained.

Variance-correction rate. Under local isotropy, the node variance satisfies $\sigma_i^2 \approx \mathbb{E}\|x - w_i\|^2$ and a radial shift δr changes the variance by $\Delta \sigma_i^2 \approx 2\sqrt{\sigma_i^2} \delta r$. Spreading this correction over $H = k$ effective updates gives the GNG learning rate

$$\eta_{\text{GNG},i} \approx \frac{\Delta \sigma_i^2}{2H \sigma_i^2} = \frac{1}{2k} \left(1 - \frac{\sigma_i^2}{\tau}\right), \quad \text{or, with the half-life convention,} \quad \eta_{\text{GNG},i} = \frac{\ln 2}{2k} \left(1 - \frac{\sigma_i^2}{\tau}\right).$$

Centering rate. The deferred nudge fires when $\|\mathbf{a}_i\|$ exceeds the resolution threshold

$$\delta_{\min} = \kappa(1 - r)\sqrt{\tau}, \quad \kappa = 0.5 \text{ (default),}$$

where $r = 1/\sqrt{2}$ is the grid ratio of S2.1. Under expected drift per update of order $\sqrt{\tau}$, the centering rate that produces one realized move per k updates satisfies $k \eta_{\text{cent}} \sqrt{\tau} = \delta_{\min}$, hence

$$\eta_{\text{cent}} = \frac{\kappa(1 - r)}{k}.$$

This couples the centering rate to the same neighborhood window used for the EWMA half-life and to the grid spacing of S2.1, so changing k or r keeps the effective dynamics scale-consistent.

S2.3.1 Variational Interpretation and Refinement Admissibility

Latent density and mesh family. Let p be the data-generating density on $\mathbb{R}^d$. The Stage 1 scaffold defines a parametric mixture

$$q_\theta(\mathbf{x}) = \sum_i r_i(\mathbf{x}; \theta) \phi_i(\mathbf{x}; \theta), \quad \phi_i = \mathcal{N}(\mathbf{w}_i, \Sigma_i),$$

with r_i the routing kernel and $\theta = \{\mathbf{w}_i, \Sigma_i\}_i$. The routing kernel approximates a Gaussian neighborhood kernel centered at each node: in low ambient dimension ($d \leq 8$) the implementation uses the Gaussian-relative weights of S2.2 directly, while for higher dimensions a rank-decay approximation preserves the same qualitative shape at lower cost. In either case, $r_i(\mathbf{x}; \theta)$ is large when $\mathbf{x}$ is close to $\mathbf{w}_i$ relative to competing nodes and decays smoothly with distance, so the induced soft assignment $\rho_i(\mathbf{x}) = r_i / \sum_j r_j$ partitions $\mathbb{R}^d$ into overlapping *catchments* whose boundaries are determined by the node geometry and the kernel bandwidth (set implicitly by τ through S2.4). Stage 1 minimizes the forward KL divergence

$$\mathcal{F}(\theta) = \text{KL}(p \parallel q_\theta) = \text{const} - \mathbb{E}_p[\log q_\theta(\mathbf{x})].$$

Stage 1 does not perform an explicit $\nabla_\theta \mathcal{F}$ step; rather, the variance-cap projection, EWMA moment tracking, and split/prune dynamics implicitly reduce $\mathcal{F}$ by keeping each component's catchment-conditional variance within the admissible family.

Catchment-conditional density. The catchment mass is $\pi_i = \int \rho_i(\mathbf{x}) p(\mathbf{x}) d\mathbf{x}$ and the catchment-conditional density is

$$p_i(\mathbf{x}) = \frac{\rho_i(\mathbf{x}) p(\mathbf{x})}{\pi_i}.$$

In EWMA equilibrium the per-node statistics recover the catchment-conditional moments:

$$\mathbf{m}_i = \mathbb{E}_{p_i}[\mathbf{x} - \mathbf{w}_i], \quad \mathbf{s}_i = \mathbb{E}_{p_i}[(\mathbf{x} - \mathbf{w}_i)^{\odot 2}], \quad \sigma_i^2 = \text{tr}(\mathbf{s}_i - \mathbf{m}_i \mathbf{m}_i^\top).$$

Thus σ_i^2 is the trace of the catchment-conditional residual covariance about the current reference $\mathbf{w}_i$, and h_i is a Monte-Carlo estimator of the catchment mass π_i scaled by the total sample count.

Hard-Voronoi variance, soft-routing mean. The variance estimator σ_i^2 and its partition-aligned shadows ($\mathbf{s}_i^\pm$) update only on BMU events (rank 0 in the k -NN result), so σ_i^2 measures the hard-Voronoi catchment variance. The residual mean $\mathbf{m}_i$ (and shadow means $\mathbf{m}_i^\pm$) update under the full soft routing kernel for all k neighbors. This asymmetry is an instance of the refinement-admissibility principle applied at the routing step: the variance cap is a geometric resolution constraint on each Voronoi cell and should measure only cell-local residuals, while the residual mean is a gradient summary used for centering and Oja direction tracking and benefits from the neighborhood smoothing of the soft kernel. Using the soft kernel for the variance would inflate σ_i^2 by a k - and kernel-dependent factor (residuals from neighboring catchments are large), making the cap comparison dataset-specific rather than geometric.

Variance cap as a resolution constraint. The admissible family at scale τ is

$$\mathcal{Q}_\tau = \{q_\theta : \text{tr}(\Sigma_i) \leq \tau_{\text{local},i} \text{ for every } i\}.$$

A node with $\sigma_i^2 > \tau_{\text{local},i}$ is outside $\mathcal{Q}_\tau$: its catchment-conditional variance exceeds the cap. Define the *variance surplus*

$$E_i = (\sigma_i^2 - \tau_{\text{local},i})_+.$$

Under the isotropic Gaussian working model $p_i \approx \mathcal{N}(\mathbf{w}_i, \sigma_i^2 I_d)$, the per-sample KL deficit between p_i and the best τ -admissible Gaussian $\mathcal{N}(\mathbf{w}_i, \tau_{\text{local},i} I_d)$ is

$$\Delta \text{KL}_i = \frac{d_{\text{final},i}}{2} \left[\frac{\sigma_i^2}{\tau_{\text{local},i}} - 1 - \log \frac{\sigma_i^2}{\tau_{\text{local},i}} \right] + O(\|\mathbf{m}_i\|^2/\tau).$$

Near the cap ($\sigma_i^2 \approx \tau_{\text{local},i}$) the bracketed term is $\frac{1}{2}(E_i/\tau_{\text{local},i})^2 + O(E_i^3)$, so E_i is, to leading order, the per-sample evidence that the current resolution is insufficient. The accumulated error evidence

$$\mathcal{E}_i = h_i E_i$$

is the cumulative log-likelihood deficit for the catchment at scale τ , serving as the continuous-Gaussian analog of the S3.4 BDeu evidence gate.

Splits as variational refinement. A split at node i along $\mathbf{u}_i$ replaces one component with two: $\rho_i \mapsto \rho_{i'} + \rho_k$. The refined family $\mathcal{Q}'_\tau \supseteq \mathcal{Q}_\tau$, so $\mathcal{F}$ weakly decreases.

Refinement-admissibility principle. When the variational family is refined (split, recursion-child, Stage 2 entry), each component of node state behaves according to whether it is a sufficient statistic of an invariant marginal or of a partition-conditional density.

1. *Flow-conserved* (sufficient statistics of a partition-marginal that decomposes exactly under refinement): $h_i, n_{i \rightarrow j}$. These are distributed between the children by the S4.3 flow-conservation rule; their sum is invariant (exactly, under the convention that partition-plane events—measure zero—are routed deterministically; cf. S2.3.2).
2. *Frame-relative* (sufficient statistics conditioned on the catchment partition): $\mathbf{m}_i, \mathbf{s}_i, \sigma_i^2, \mathbf{a}_i$, and the EWMA precision counter (update count). The pre-split aggregate moments are sufficient for the parent catchment p_i , but the mapping to the post-split child catchments $p_{i'}$, p_k is non-identifiable from the aggregates alone because the per-sample partition membership was not stored. The prior-consistent baseline is to reset these to the EWMA prior (zero), unless the implementation maintains partition-aligned shadow moments (S2.3.2) that recover identifiability (see S2.3.2 for the streaming workaround).
3. *Structural / geometric* (parameters of the family itself): $\mathbf{w}_i, \mathbf{u}_i, d_{\text{final},i}$. The parent keeps $\mathbf{w}_i$; the child inherits $\mathbf{u}_i$ and $d_{\text{final},i}$.

Connection to cross-frame transfers. The three warm-start regimes of S4.4 are instances of this principle: T1 cross-grid applies a partial shrink $\gamma_0 \approx 0.7$ because the catchment is similar but the scale has changed; T2 parent-to-child applies a full reset because the partition changes drastically; T3 Stage 1-to-Stage 2 applies a partial shrink because the catchment is unchanged but the model class shifts. The S12.6 EWMA noise bound is well-posed precisely because the splitting node and its child enter the post-split trajectory at the EWMA prior (or at the shadow-inherited precision), rather than at the parent’s stale over-cap value.

S2.3.2 Partition-Aligned Shadow Moments

The moment-reset baseline of S2.3.1 is correct but conservative: it discards evidence that the implementation can retain cheaply. If the node continuously tracks the Oja direction $\mathbf{u}_i$, a pair of *shadow EWMA*s partitioned by $\text{sign}((\mathbf{x} - \mathbf{w}_i) \cdot \mathbf{u}_i)$ recovers identifiability at the moment of a split.

Shadow state. Each node maintains, alongside the aggregate moments $(\mathbf{m}_i, \mathbf{s}_i, h_i)$, a positive-side shadow $(\mathbf{m}_i^+, \mathbf{s}_i^+, h_i^+)$ and a negative-side shadow $(\mathbf{m}_i^-, \mathbf{s}_i^-, h_i^-)$. For a routed sample $\mathbf{x}$ with residual $\mathbf{e}_i = \mathbf{x} - \mathbf{w}_i$, the aggregate update proceeds as in S2.3, and additionally:

- if $\mathbf{e}_i \cdot \mathbf{u}_i > 0$, update $(\mathbf{m}_i^+, \mathbf{s}_i^+)$ with the same rank-decay-weighted EWMA gain as the aggregate update in S2.3 and residual $\mathbf{e}_i$, and increment h_i^+ by the routing weight;
- if $\mathbf{e}_i \cdot \mathbf{u}_i < 0$, update $(\mathbf{m}_i^-, \mathbf{s}_i^-)$ analogously and increment h_i^- ;
- if $\mathbf{e}_i \cdot \mathbf{u}_i = 0$ (measure-zero event), neither shadow is updated.

Split inheritance. When a split fires along $\mathbf{u}_i$, the positive-side child inherits $(\mathbf{m}_i^+, \mathbf{s}_i^+, h_i^+)$ as its initial aggregate moments, and the parent (now the negative-side child) inherits $(\mathbf{m}_i^-, \mathbf{s}_i^-, h_i^-)$. Both children reset their own shadow pairs to zero (they will accumulate new shadows against their respective post-split Oja directions). The deferred nudge $\mathbf{a}_i$ and the EWMA precision counter (update count) reset to zero on both children, because the nudge references the parent’s old position basin and the precision counter should reflect the child’s local EWMA age.

Hit counts are distributed as $h_{i'} = h_i^-$, $h_k = h_i^+$, satisfying flow conservation $h_{i'} + h_k \leq h_i$ (the inequality accounts for any samples that fell exactly on the partition plane). Transition counts on incident links are halved as in S4.3.

Steiner shift at inheritance. The shadow moments $(\mathbf{m}_i^\pm, \mathbf{s}_i^\pm)$ are accumulated relative to the parent position $\mathbf{w}_i$, but the children are placed at $\mathbf{w}_i \pm \delta \mathbf{u}_i$. To reference the inherited moments to the child’s new position, a parallel-axis (Steiner) correction is applied: for the positive-side child with displacement $\mathbf{d} = +\delta \mathbf{u}_i$,

$$\mathbf{m}_c = \mathbf{m}_i^+ - \mathbf{d}, \quad \mathbf{s}_c = \mathbf{s}_i^+ - 2 \mathbf{d} \odot \mathbf{m}_i^+ + \mathbf{d}^{\odot 2},$$

and analogously for the negative-side child with $\mathbf{d} = -\delta \mathbf{u}_i$. The variance $\text{tr}(\mathbf{s}_c - \mathbf{m}_c \odot \mathbf{m}_c) = \text{tr}(\mathbf{s}_i^\pm - \mathbf{m}_i^\pm \odot \mathbf{m}_i^\pm)$ is shift-invariant, so the cap-decision state is unchanged; the correction ensures that the post-split residual-mean direction is referenced to the child’s actual position rather than the parent’s old origin.

Hard-Voronoi variance conservation. Under hard Voronoi assignment, the parent cell partitions exactly into the two children’s cells at the split plane. The variance evidence is therefore conserved:

$$\sigma_i^2 h_i = \sigma_{i'}^{2,+} h_i^+ + \sigma_{i'}^{2,-} h_i^-.$$

This holds in the population limit; the EWMA tracks the running approximation within the half-life k . Combined with the shadow transfer, the entire variance budget of the parent is distributed to the children without loss or inflation—the children’s cap comparisons begin from the exact pre-split evidence rather than from a reset prior.

Identifiability. The shadow moments are sufficient statistics for the partition the split commits to, whenever $\mathbf{u}_i$ is approximately stationary over the shadow accumulation window. They are *not* non-identifiable, because the system chose to maintain them along the axis it will eventually split on. Rapid Oja rotation degrades sufficiency; the staleness reset below addresses this by falling back to the moment-reset baseline of S2.3.1. Subject to this stationarity proviso, shadow transfer is the unique streaming-compatible mechanism for preserving evidence across a refinement when only the dominant partition axis is known in advance.

Oja drift and staleness. As $\mathbf{u}_i$ rotates, the shadow partition plane drifts. The shadow EWMA uses the same half-life k as the aggregate moments, so they track the current Oja direction at the same effective rate. If $\mathbf{u}_i$ undergoes a large discrete jump (e.g. after a nearby topology edit), the shadows can be flagged stale and reset, falling back to the moment-reset baseline of S2.3.1.

Cost. Memory: one extra moment pair per node, i.e. $3 \times$ per-node moment storage; still $O(d)$ and streaming-bounded. Compute: one extra dot product and one branched EWMA update per routed sample per node; no per-sample retention.

S2.4 Control-to-Threshold Mapping

The global scale variable is the variance cap τ . A bounded controller value $s_{\text{control}} \in (0, 1)$ is mapped to a variance threshold by

$$\tau = -D_{\text{subspace}} \log(1 - s_{\text{control}}),$$

where D_{subspace} is the working dimensionality of the current region’s coordinate frame (ambient dimension at the root; the retained PCA rank from S1.4 at child recursion levels). Within a single scaffold run, all nodes share the same cap: $\tau_{\text{local},i} = \tau$ for every i . Dimensional heterogeneity within a region is resolved by the scale-response machinery (S2.5–S2.6) peaking at different τ for clusters of different intrinsic dimension, and by recursion giving each cluster its own scale search after PCA transfer. The per-node intrinsic-dimension estimate $d_{\text{final},i}$ is retained as a diagnostic input to junction detection (S8.4) and as the cluster-median aggregation target when a single d_C is required for scale-response normalization, but it does not modulate the split-decision cap. Stationarity at a given τ is measured by $\text{CV}(\sigma_i^2)$ over mature nodes; since τ is uniform, equilibrium means uniform-resolution evidence across the scaffold.

S2.5 Scale Response $\Phi_C(\tau)$

For a stabilized scaffold at scale τ , define a per-node Lindeberg-normalized response $R_i(\tau)$ from the node’s local hit mass, neighborhood volume proxy, and intrinsic dimension estimate. Let h_i be node i ’s fractional hit mass, $r_{k,i}$ its distance to the k -th nearest scaffold neighbor, N_C the number of routed samples in the current region, d the local working dimension, and V_d the volume of the unit d -ball. The standard k -NN density proxy is

$$\hat{\rho}_i = \frac{h_i k N_C}{V_d r_{k,i}^d}.$$

With the scale calibration $\sigma = \sqrt{\tau}/c_{d,k}$ from S11, the Lindeberg-normalized node response is

$$R_i(\tau) = \left(\frac{\sqrt{\tau}}{c_{d,k}} \right)^d \hat{\rho}_i = \left(\frac{\sqrt{\tau}}{c_{d,k}} \right)^d \frac{h_i k N_C}{V_d r_{k,i}^d}.$$

The cluster response is

$$\Phi_C(\tau) = \sum_{i \in C} R_i(\tau).$$

The sum is the scale-selection object: it detects when the region’s support is well resolved at the current variance cap. In an equilibrated cluster, node hit masses are approximately equal, so individual $R_i(\tau)$ values are not reliable within-cluster density indicators. Density and membership later come from the spatial distribution of nodes, per-node residual moments, PL density reconstruction, and Gaussian summaries.

Using the same $r_{k,i}$ values, define the local support proxy

$$\hat{V}_i = \frac{V_d r_{k,i}^d}{k/N_C}, \quad V_C(\tau) = \sum_{i \in C} \hat{V}_i.$$

The support trace $V_C(\tau)$ is a cheap secondary diagnostic: robust characteristic scales should show a response peak in Φ_C near a plateau transition in V_C .

S2.5.1 Scale Controller Procedure

Evaluate $\Phi_C(\tau)$ on a geometric grid in $\log \tau$. Candidate peaks are detected by sign changes in the centered second difference, then refined with a small number of additional evaluations near each bracket. A support trace, such as a summed local volume proxy $V_C(\tau)$, is logged as a cross-check: robust characteristic scales should coincide with the expected transition between unresolved support and redundant over-refinement. A grid point is eligible only if its Stage 1 scaffold stabilizes under the fixed-scale criteria. If no robust peak exists, the region is terminal for the searched band.

The fixed- τ stopping rule should be confidence-based rather than a pure absolute-CV threshold. The minimal criterion has two layers:

1. *Topology quiet*: over a trailing window of W epochs, there are no accepted splits, no accepted node prunes, node count is unchanged, and the count of lifted isolated mature nodes is not increasing.
2. *Moment flatness*: over the same window, the slope of $\text{CV}(\sigma_i^2)$ is near zero, its rolling standard deviation is small relative to its mean, reconstruction error is no longer trending, and the mean load $\mathbb{E}[\sigma_i^2/\tau]$ is flat rather than still drifting toward the cap.

This replaces the weaker rule "the last few CV values fall below a tolerance." Absolute thresholds remain useful as sanity bounds, but the primary stopping signal is *trend exhaustion*: the scaffold has become structurally quiet and its moments are no longer evolving systematically. Optional frozen-topology epochs may still be used as a diagnostic hardening step, but they are not the default equilibrium criterion.

S2.5.2 Local Noise Estimation from Settled Catchments

The same local asymptotic regime that justifies treating a mature catchment as approximately affine also makes it a probe of the local noise field. Write a sample near x_0 as

$$X = x_0 + Uz + \frac{1}{2}\text{II}(z, z) + \varepsilon,$$

where Uz is the local tangent contribution, $\text{II}(z, z)$ is the curvature correction, and ε is a local noise term with covariance $\Psi(x_0)$. For a settled single-sheet catchment of effective radius h ,

$$\Sigma_{\text{catchment}} \approx \Psi(x_0) + U\Sigma_z(h)U^\top + \Sigma_{\text{curv}}(h) + O(h\|\nabla\Psi\|).$$

Tangential signal variance scales like $O(h^2)$, curvature corrections are higher order, and floor variation enters through the final term. As h shrinks inside a settled region, the catchment covariance therefore approaches a patchwise noise tensor plus a controlled geometric term. The practical claim is not that Proteus recovers a full covariance field everywhere, but that settled catchments provide the right local statistics from which conservative floor estimates can be extracted.

This is the bridge between scaffold settlement and local noise estimation: equilibrium is valuable not only because it stabilizes routing, but because it produces a region in which residual covariance can be interpreted as local floor evidence rather than as transient topology noise.

S2.5.3 Minimal Viable Noise Atlas

The publishable baseline is intentionally small. Proteus does *not* require a full hierarchy of floor models to operationalize local noise. The minimal atlas consists of three derived quantities:

1. **Node-level floor proxy** ψ_i . A scalar local residual scale derived from settled node moments, e.g. a trailing or robust minimum component of the residual covariance proxy.
2. **Region-level floor summary** u_{local} . A robust region aggregate of the node-level proxies, such as a median or trimmed mean over mature nodes in the region.
3. **Parent-child residual refinement**. At recursion events, refine the parent’s floor summary by subtracting child-explained covariance from the parent’s total covariance, yielding a conservative unexplained residual floor.

All three are confidence-gated: low-maturity nodes, unstable catchments, or disagreeing local summaries reduce confidence and suppress aggressive use of the floor estimate in controller decisions. This minimal atlas is sufficient to support local-floor-relative detectability, adaptive scale selection, and recursion timing without committing the first paper to a full anisotropic covariance-learning framework.

S2.6.1 Cluster Identification, Q-Score, and Cross-Scale Tracking

At a fixed variance scale v , define the effective correlation radius

$$\ell(v) = \sqrt{d_{\text{eff}} v}.$$

The scaffold graph at this scale carries edge evidence combining local geometry and routed co-activation. For a candidate node set S , define the mean internal evidence $\text{LocalIntra}(S; v)$ and mean boundary evidence $\text{BoundaryInter}(S; v)$, and set

$$Q(S; v) := \log \frac{\text{LocalIntra}(S; v) + \varepsilon}{\text{BoundaryInter}(S; v) + \varepsilon}.$$

The graph-local cluster definition used in Stage 1 is:

1. *Coherence*: $Q(S; v) > 0$;
2. *Extent*: $\hat{v}_S \geq v$, so the set is not sub-resolution at the current operating scale;
3. *Maximality*: no adjacent candidate T has $\text{InterLocal}(S, T; v)$ as large as either region’s internal evidence.

For a partition $\mathcal{P} = \{S_1, \dots, S_k\}$, the region-weighted quality is

$$Q(\mathcal{P}; v) = \sum_{j=1}^k \frac{|S_j|}{n} Q(S_j; v).$$

Merges are admissible only when the merged partition increases $Q(\mathcal{P}; v)$ and the inter-cluster coupling is no weaker than the weaker of the two intra-cluster coherences; split proposals are admissible only when every child still satisfies coherence and extent at the current scale.

At the selected scale τ_C^* , Stage 1 uses Affinity Propagation on the stabilized Hebbian graph to produce candidate exemplars and fragments. Similarities are computed only on graph edges using smoothed transition probabilities and a smoothed-PMI score. For directed Hebbian counter $C(i \rightarrow j)$, define

$$\hat{q}(j \mid i) = \frac{\alpha_{0,i} + C(i \rightarrow j)}{\sum_{\ell \in \mathcal{N}(i)} (\alpha_{0,i} + C(i \rightarrow \ell))}.$$

Let $K_i = |\mathcal{N}(i)|$ and define the smoothed marginal node mass

$$\hat{P}(i) = \frac{h_i + \alpha_{0,i}K_i}{\sum_v (h_v + \alpha_{0,v}K_v)}.$$

For an undirected Hebbian edge $\{i, j\}$, use the symmetrized joint estimate

$$\hat{P}(i, j) = \frac{1}{2} \left(\hat{P}(i)\hat{q}(j | i) + \hat{P}(j)\hat{q}(i | j) \right),$$

and set the AP similarity to smoothed PMI,

$$s(i, j) = \log \hat{P}(i, j) - \log \hat{P}(i) - \log \hat{P}(j).$$

No similarity is assigned to unlinked node pairs. The AP self-preference is derived from the same BDeu concentration:

$$s(i, i) = -2\log(d_{\text{final},i} + 1).$$

This makes low-dimensional regions more willing to instantiate exemplars and high-dimensional regions more conservative.

AP exemplars are discrete representatives for graph clustering; they are not the continuous modes used in fuzzy membership. The directional information discarded by symmetrization is retained as a diagnostic

$$A(i, j) = \frac{C(i \rightarrow j) - C(j \rightarrow i)}{C(i \rightarrow j) + C(j \rightarrow i) + \varepsilon}.$$

$A(i, j)$ is not used for AP clustering; it is logged for dimensionality-junction diagnostics. Across adjacent grid values, clusters can be tracked by node-ID overlap because the scaffold is warm-started. After selection, each AP cluster becomes a child region, and samples route to the child owning their BMU.

AP fragments are only proposals. Stage 1 subsequently performs Q-gated pair merges, boundary refinement, and residual cleanup so that the accepted partition satisfies the graph-local cluster definition rather than merely the AP objective.

S2.6.2 Feature Persistence and Recursion Timing

Every scale-space feature has a persistence interval: the range of scales over which it satisfies the cluster definition. There are two distinct cases.

- **Intrinsic features:** true features of the unsmoothed density ρ itself. They have a coarse-end death scale v_{death}^+ above which smoothing destroys their contrast or merges them into a neighbor. By Gaussian non-enhancement they persist monotonically toward finer scales.
- **Composite features:** features of the smoothed family ρ_v that summarize unresolved children over an intermediate range of scales. They have both a coarse-end death scale v_{death}^+ and a fine-end death scale v_{death}^- below which they resolve into children.

For a feature F_k , the useful characteristic scales are $v_{\text{death},k}^+$, v_k^* , σ_k^2 , $v_{\text{death},k}^-$, and $\tau_{\text{sep},kj}$. Here v_k^* is the preferred operating scale, σ_k^2 is the intrinsic width, $v_{\text{death},k}^-$ is defined only for composites, and $\tau_{\text{sep},kj}$ is the scale at which a feature first becomes distinguishable from neighbor j . Traversing from coarse to fine, a well-posed feature satisfies

$$v_{\text{death},k}^+ > v_k^* \gtrsim \sigma_k^2 > v_{\text{death},k}^-.$$

For intrinsic features, $v_{\text{death},k}^- = 0$.

This yields the recursion-timing interpretation used by Stage 1. The discovery objective is coarse-to-fine: identify a feature as close to v_k^* as possible without overshooting its coarse-end

death. For composites, parent fine-end death and child coarse-end birth are dual signals of the same event. In the clean handoff regime,

$$v_{\text{death,parent}}^- \approx \max_j v_{\text{death,child}_j}^+.$$

The parent begins to fail coherence or maximality at roughly the same scale at which its strongest child becomes independently detectable. Monitoring both signals makes recursion predictive rather than purely reactive.

S2.7 Dirichlet Concentration from the BDeu Prior

Let J_i be the effective number of outgoing outcomes for node i , approximated by the local branching factor $d_{\text{final},i} + 1$. The BDeu convention assigns one equivalent pseudo-observation spread uniformly over these outcomes:

$$\alpha_{0,i} = \frac{1}{d_{\text{final},i} + 1}.$$

This concentration is used consistently in transition smoothing, AP self-preference, and the Dirichlet–multinomial evidence gate. The point is not to tune a separate smoothing knob for every dimension; the same dimension-aware prior regularizes zero counts and supplies the Occam factor in structure selection.

S2.8 Resolution Theory

The graph-local Q-score of S2.6 is the algorithmic separability criterion. This subsection states the corresponding information-theoretic preconditions under which the cluster definition can be satisfied at all.

Density model. At fixed local effective dimension d , model the density as

$$\rho(x) = u + \sum_{i=1}^m H_i \exp\left(-\frac{\|x - \mu_i\|^2}{2\sigma_i^2}\right),$$

where $u > 0$ is the local tissue floor, H_i is the peak excess above tissue, σ_i is the intrinsic width, and μ_i is the mode center. Under Gaussian smoothing by variance τ , a single mode evolves as

$$\rho_\tau(x) = u + H_i \left(\frac{\sigma_i^2}{\sigma_i^2 + \tau}\right)^{d/2} \exp\left(-\frac{\|x - \mu_i\|^2}{2(\sigma_i^2 + \tau)}\right).$$

The smoothed width and peak excess are therefore

$$s_i(\tau) = \sqrt{\sigma_i^2 + \tau}, \quad h_i(\tau) = H_i \left(\frac{\sigma_i^2}{\sigma_i^2 + \tau}\right)^{d/2}.$$

Detectability above tissue. Let u denote the local floor relevant to the current recursion context. Define the scale-dependent signal-to-tissue contrast

$$C_i(\tau) := \frac{h_i(\tau)}{u} = \frac{H_i}{u} \left(\frac{\sigma_i^2}{\sigma_i^2 + \tau}\right)^{d/2}.$$

For a mode to satisfy the coherence part of the cluster definition, it must retain enough excess above this floor:

$$C_i(\tau) \geq \kappa_{\text{det}}.$$

Equivalently,

$$u \leq \frac{H_i}{\kappa_{\text{det}}} \left(\frac{\sigma_i^2}{\sigma_i^2 + \tau} \right)^{d/2},$$

and the maximum admissible operating scale is

$$\tau_{\text{det},i} := \sigma_i^2 \left[\left(\frac{H_i}{\kappa_{\text{det}} u} \right)^{2/d} - 1 \right].$$

At the natural operating scale $\tau = \sigma_i^2$, this becomes the characteristic-scale detectability condition

$$\frac{H_i}{u} \geq \kappa_{\text{det}} 2^{d/2}.$$

Clean resolvability of two modes. Two equal-width isotropic Gaussian modes of common width σ , separated by Δ , remain bimodal after smoothing by variance τ if and only if

$$\Delta > 2\sqrt{\sigma^2 + \tau}.$$

The associated maximum separation-preserving operating scale is

$$\tau_{\text{sep}} = \frac{\Delta^2}{4} - \sigma^2.$$

At the natural operating scale $\tau = \sigma^2$, the characteristic-scale threshold is

$$\Delta > 2\sqrt{2}\sigma.$$

For unequal widths, a conservative sufficient condition is

$$\Delta > 2 \max(\sqrt{\sigma_a^2 + \tau}, \sqrt{\sigma_b^2 + \tau}).$$

Bridge to the graph-local Q criterion. If two clusters are locally Gaussian with empirical scales $\hat{v}_A, \hat{v}_B$ and centroid separation Δ_{AB} , their all-pairs inter-cluster coupling decays approximately as

$$\text{Inter}(A, B; v) \approx \exp\left(-\frac{\Delta_{AB}^2}{2 d_{\text{eff}}(v + \hat{v}_A + \hat{v}_B)}\right).$$

Clean graph-level separation requires this inter-cluster term to be much smaller than typical intra-cluster evidence, yielding the heuristic condition

$$\Delta_{AB}^2 \gg d_{\text{eff}}(v + \hat{v}_A + \hat{v}_B).$$

At equilibrium, where $v \approx \hat{v}_A \approx \hat{v}_B \approx \sigma^2$, this becomes

$$\Delta_{AB}^2 \gg 3 d_{\text{eff}} \sigma^2.$$

The continuous bimodality bound is therefore the conservative outer envelope, while the Q-gated scaffold may sometimes resolve slightly tighter pairs because it also uses co-activation evidence. The gap between the continuous bound and the graph-local bound is the operating margin.

S3. Structure Selection and Statistical Gates

S3.1 Link Pruning

A new link is protected until it has enough effective observations for a meaningful comparison: $n_{\text{eff}} \geq n_{\text{req}}$ from the analytic power calculation for proportions. After this shield expires, compute a one-sided Wilson upper bound for the link's relative transition proportion p from count n :

$$\nu^+ = \frac{p + z^2/(2n) + z\sqrt{p(1-p)/n + z^2/(4n^2)}}{1 + z^2/n}, \quad z = 1.2816$$

corresponding to a 90% one-sided comparison. Vote to prune when ν^+ is below the median peer significance in the same local star, and require bilateral agreement from both endpoints. This prevents transient low counts from deleting bridges before they have had a fair chance to accumulate evidence.

S3.2 Node Pruning

A node is a prune candidate only if it is both statistically weak and geometrically unneeded. The default rule combines:

- neighborhood stability, using a Welch test on local density/response estimates;
- an arrival-rate fairness test $\lambda_{\text{est}} = h_i/\text{lifetime} \geq 2/\delta_{\min}$;
- low relative hit mass, $h_i < 0.5$ times the neighbor mean;
- low local variance, $\sigma_i^2 < 0.5 \tau_{\text{local},i}$.

Low hits alone are not sufficient: a low-mass node in a high-variance or curved region may be structurally important and should not be removed until the region has stabilized.

S3.3 Stage 2 Prunes and Merges

Stage 2 node deletions are arbitrated by the node's simplex star. A deletion proposal is vetoed if removing the node creates a large local reconstruction error, destabilizes adjacent stars, or violates the local reassignment radius bound. Prune candidates inherit $h_j < 0.5\bar{h}$ and $\sigma_j^2 < 0.5 \tau_{\text{local},j}$ from S3.2 and additionally require

$$\rho_{\text{prune}} := \max_{x \in V_j} \min_{s \neq j} \|x - w_s\| \leq C_Q(d) \sqrt{\tau_{\text{local},j}},$$

where $C_Q(d)$ is the variance-cap star-radius constant in the regular interior. Merge proposals $\{a, b\} \mapsto m$ require, before evidence scoring,

$$\|w_a - w_b\| \leq \rho_{\text{merge}} \sqrt{\tau_{ab}}, \quad P(V_a \cup V_b) \leq p_{\text{merge}}, \quad \max(\sigma_a^2, \sigma_b^2) \leq v_{\text{merge}} \tau_{ab},$$

with defaults

$$\rho_{\text{merge}} \leq 2C_Q(d), \quad p_{\text{merge}} \leq \bar{h}/N, \quad v_{\text{merge}} \leq 1, \quad \tau_{ab} = \max(\tau_{\text{local},a}, \tau_{\text{local},b}).$$

Under these guards, the per-edit positive vertex-potential jump is bounded by

$$\Delta L_{\text{vertex}}^{\text{prune/merge}} \leq B_p = \max\{p_{\text{prune}}(\rho_{\text{prune}}^2 + 2\rho_{\text{prune}}\sqrt{v_{\text{prune}}\tau_*}), p_{\text{merge}}(\rho_{\text{merge}}^2 + 2\rho_{\text{merge}}\sqrt{v_{\text{merge}}\tau_*})\},$$

with $\tau_* = \max_i \tau_{\text{local},i}$ and global bounded-support fallback $B_p \leq R^2$.

S3.4 Dirichlet–Multinomial Evidence Score

For candidate topology M , define the affected node set $\mathcal{V}_{\text{aff}}$ as the nodes whose stars or neighbor sets differ between the candidate and keep-topology baselines, plus a neighbor ring. Let $J_i(M)$ be the number of outgoing categorical outcomes at node i under M . With BDeu concentration $\alpha_{0,i}$, the node-local Dirichlet–multinomial marginal is

$$p_i(M) = \frac{\Gamma(J_i \alpha_{0,i})}{\Gamma(J_i \alpha_{0,i} + n_i)} \prod_{j=1}^{J_i} \frac{\Gamma(\alpha_{0,i} + n_{i \rightarrow j})}{\Gamma(\alpha_{0,i})}.$$

The affected-region score is

$$F_{\text{DM}}(\mathcal{R}; M) = - \sum_{i \in \mathcal{V}_{\text{aff}}} \log p_i(M).$$

A candidate is accepted only if

$$F_{\text{DM}}(\mathcal{R}; M_{\text{edit}}) < F_{\text{DM}}(\mathcal{R}; M_{\text{keep}}) - \log \tau_{\text{BF}},$$

where $\log \tau_{\text{BF}}$ is the log-Bayes-factor margin. Candidate-dependent preprocessing is limited to deterministic rerouting of affected events under the candidate topology.

S3.5 Evidence Consistency

For any finite candidate set $\mathcal{M}_{\mathcal{R}}$, assume the induced routers are identifiable and regular, routed transition events are i.i.d. or sufficiently mixing, probabilities are clipped below by $q_{\min} > 0$ for finite-sample bounds, and the best candidate has positive expected log-likelihood margin $\Delta_{\mathcal{R}} > 0$. Then the Dirichlet–multinomial marginal selects the best candidate with probability tending to one as the affected-region event count grows. The likelihood margin scales as $O(H_{\mathcal{R}} \Delta_{\mathcal{R}})$, while the BIC/Occam and fixed-threshold terms grow only logarithmically or remain constant.

S3.6 Gate Cadence and Edit Budgets

Proposals are appended to a per-region queue $\mathcal{Q}_C$ keyed by diagnostic strength. The gate fires after an equilibration window

$$W = \max\left(N_{\text{nodes}}^{(C)}, 4k |\mathcal{Q}_C|\right)$$

samples, which gives each affected node several updates before scoring; the window terminates early if $\mathcal{Q}_C$ is empty. Disjoint proposals may be scored in parallel. Overlapping proposals are processed in priority order, and proposals whose affected region overlaps an accepted edit are marked stale and re-evaluated against the post-edit topology.

Accepted edits trigger a short local settle of order $|\mathcal{R}|k$ updates. Hysteresis prevents immediate reversals: an edit cannot be reverted within

$$T_{\text{hyst}} = 2W$$

samples of acceptance. A per-level edit budget

$$N_{\text{edit}}^+(C) \leq \frac{N_{\text{nodes}}^{(C)}}{\log N_{\text{nodes}}^{(C)}}$$

accepted prune/merge edits per training epoch bounds the cumulative positive jump:

$$A_{\text{epoch}}^{(C)} \leq B_p N_{\text{edit}}^+(C) \leq B_p \frac{N_{\text{nodes}}^{(C)}}{\log N_{\text{nodes}}^{(C)}}.$$

For finite-epoch training, $A_T < \infty$ over each epoch. Indefinite streaming is treated as an engineering extension whose edit-rate tolerance is monitored rather than claimed as a first-paper guarantee.

S4. Stage 2 Complex Construction

S4.1 Flag-Complex Invariant

For a fixed region C , Stage 2 represents topology as the top-dimensional flag complex of the current Hebbian graph. A d -simplex $\sigma = \{v_0, \dots, v_d\}$ is present exactly when every pair of vertices in σ is a Hebbian edge and the local intrinsic-dimension rule permits a d -simplex in that star. Node and link edits therefore induce simplex edits automatically: deleting a link deletes every simplex containing that link, and deleting a node deletes its full star. Geometric motion alone does not change simplex membership.

This invariant avoids separate Delaunay or alpha-complex maintenance. The graph owns topology; the complex is the closure of that topology at the locally estimated dimension. Geometry-dependent caches are recomputed only when the affected simplex geometry changes.

S4.2 Greedy Chaining Initialization

Greedy Chaining initializes the flag complex from the stabilized Stage 1 graph. For each seed node, inspect rank-ordered neighbor subsets at the local intrinsic dimension, verify clique completion with adjacency hash lookups, and insert the canonical sorted simplex into a deduplication set. A breadth-first traversal over the graph amortizes the search because many candidate simplices are encountered from multiple seeds.

For heterogeneous intrinsic dimension, node i enumerates subsets sized by $d_{\text{final},i}$. Cross-junction candidates that do not form valid cliques simply fail verification; the resulting incidence-count distribution is itself a useful junction diagnostic. Nodes with zero simplex incidence after the main pass receive one final node-seeded extension attempt. Remaining orphans are deferred to continuous simplex discovery when later Hebbian links complete valid cliques.

S4.3 Flow-Conserving Split Initialization

Flow conservation is the special case of the refinement-admissibility principle (S2.3.1) for sufficient statistics of partition-marginal flux: hit counts and transition tallies decompose exactly under refinement because the catchment mass partitions, so the children's totals must sum to the parent's.

When a split inserts a new vertex k into a parent simplex or node star, transition tallies should be initialized so the local flow field is not artificially reset. If C is a parent simplex with vertices $V(C)$ and $h_{\text{old}}(v_i, v_j)$ denotes pre-split internal hit counts, initialize new tallies by distributing the old internal flow through the new vertex:

$$h_{\text{new}}(k, v_i) = \frac{1}{2} \sum_{\substack{v_j \in V(C) \\ j \neq i}} h_{\text{old}}(v_i, v_j).$$

Simultaneously, original internal edges are halved: $h_{\text{old}}(v_i, v_j) \rightarrow \frac{1}{2} h_{\text{old}}(v_i, v_j)$ for all $v_i, v_j \in V(C)$ with $v_i, v_j \neq k$. Under this convention total internal mass is preserved exactly:

$$\sum_i h_{\text{new}}(k, v_i) + \sum_{\substack{i < j \\ v_i, v_j \neq k}} \frac{1}{2} h_{\text{old}}(v_i, v_j) = \sum_{i < j} h_{\text{old}}(v_i, v_j).$$

The rule matters for dynamic stability: accepted edits should perturb routing statistics by a bounded amount rather than erasing local evidence.

S4.4 Transition Counts Across Levels and Stages

There are four warm-start regimes, each an instance of the refinement-admissibility principle of S2.3.1.

- T0. Within-region split.* A variance-cap split along $\mathbf{u}_i$ produces two children from one node. Each child inherits the shadow pair corresponding to its side of the split plane (S2.3.2), with a Steiner parallel-axis correction so the inherited moments are referenced to the child’s new position rather than the parent’s old origin. Both children reset their own shadow pairs, deferred nudge $\mathbf{a}_i$, and EWMA precision counter to zero, and inherit $\mathbf{u}_i$ and $d_{\text{final},i}$. Hit counts distribute by shadow masses $h_i^\pm$, satisfying hard-Voronoi conservation. Transition counts on incident links are halved per S4.3. If shadows are not maintained, both children reset all frame-relative state to the EWMA prior (zero) and split hit counts evenly.
1. *T1: cross-grid within one recursion level.* Adjacent scale-grid evaluations preserve topology and neighbor sets. Hit counts h_i and transition counts $n_{i \rightarrow j}$ are inherited unchanged; EWMA moments $(\mathbf{m}_i, \mathbf{s}_i)$ are rescaled or attenuated using the same default shrink $\gamma_0 \approx 0.7$ used at Stage 2 entry; the nudge accumulator $\mathbf{a}_i$ is reset to zero.
 2. *T2: parent-to-child recursion.* A child region restricts the neighbor set and sample support, then applies the S1.4 intrinsic-dimension PCA transfer in the active parent frame. Child samples and inherited node positions are projected into the child PCA frame, D_{subspace} is set to the retained child rank, and inherited d_{final} values carry forward as the dimension seeds for the child. Hit counts, transition counts, EWMA moments, and nudges are reset to zero. This restores the BDeu/DM interpretation under the child’s restricted router.
 3. *T3: Stage 1 to Stage 2.* Stage 2 inherits stabilized node positions, links, hit counts, and transition counts. EWMA moments are shrunk by $\gamma_0 \approx 0.7$, and the nudge accumulator is reset to zero.

These policies are part of the statistical specification, not implementation details: they determine when counts remain evidence for the same router and when they must be re-measured.

S4.5 Stage 2 Data Structures

Each simplex maintains the small geometry and tally caches needed by later stages: vertex IDs, facet IDs, volume $|S|_d$, face normals, pairwise edge norms, face-pressure tallies (`face_pressures`, size $d+1$), torsion accumulators (`torsion_sum`, `torsion_sum_sq` for κ_S), and a `simplex_update_count`. The implementation does not cache per-simplex shape-quality metrics such as inradius/circumradius ratios. ANN indices are maintained at the node level and refreshed after deferred moves or topology edits that affect local routing.

Rank-1 Oja vs. scalar variance. The core convergence theory holds with a scalar variance tracker, but the recommended implementation augments it with a rank-1 Oja update of the dominant local-residual direction $\mathbf{u}_i$. This adds one stored vector per node and gives a more accurate placement direction for proposed splits. For ambient dimensions $d \lesssim 10,000$ the memory cost is modest and the split-quality gain justifies it; for very high-dimensional data, fall back to the scalar-only variance with random or Oja-projection-based split direction.

Numerical-stability constants. Recommended clipping defaults:

- Set ε in $\rho_i = \|\mathbf{m}_i\|/(\sigma_i + \varepsilon)$ to between 10^{-8} and 10^{-6} depending on data scale.
- Clip the GNG-style learning rate to $|\eta_{\text{GNG}}| \leq 0.3$ in Stage 1 and $|\eta_{\text{GNG}}| \leq 0.05$ in Stage 2 (where $k \approx d$ makes the per-update displacement larger).

- Constrain spline-flow slopes/bins to avoid near-singular Jacobians during warp fits.
- Bound $\|\mathbf{a}_i\|$ growth between deferred-nudge applications, so a single sample cannot push the accumulator above several $\delta_{\min}$.

HNSW defaults: graph degree $M \in [16, 32]$, efConstruction $\in [100, 400]$, efSearch $\in [50, 200]$. Increase efSearch modestly near convergence to improve neighbor stability. Persist indices per recursion level and rebuild only after significant mesh mutation.

S4.6 Numerical Cost Envelopes

Stage 1 cost is dominated by ANN queries and rank-weighted local updates. Greedy Chaining is local: each accepted simplex pays clique verification and a canonical deduplication insert, with traversal amortized over shared stars. Evidence scoring is local in the affected region and scales with the number of changed stars plus a neighbor ring. Torsion audits and edge-ratio split checks use cached simplex-local matrices and pairwise edge norms. Dual-flow reconstruction runs on the face/dual graph of the accepted complex. These envelopes should be logged by owned object (nodes, edges, simplices, faces, proposals) so scaling bottlenecks can be diagnosed without mixing stages.

S5. Torsion, Safe Edits, and Warps

S5.1 Discrete Torsion Diagnostic

For a d -simplex $S = \{v_0, \dots, v_d\}$, let $\bar{\mathbf{w}}$ and $\bar{\mathbf{m}}$ be the barycenters of vertex positions and residual means. Form centered matrices

$$E = [\mathbf{w}_{v_1} - \bar{\mathbf{w}} \ \cdots \ \mathbf{w}_{v_d} - \bar{\mathbf{w}}], \quad M = [\mathbf{m}_{v_1} - \bar{\mathbf{m}} \ \cdots \ \mathbf{m}_{v_d} - \bar{\mathbf{m}}].$$

The simplex torsion form is

$$\Omega_S = M^\top E - E^\top M, \quad \kappa_S = \|\Omega_S\|_F, \quad R_S = \kappa_S / \tau^*.$$

If the residual mean field is locally consistent with the gradient of a potential, the implied Jacobian is approximately symmetric and Ω_S is small. Persistent high R_S indicates that local PL geometry is not explaining the residual field conservatively.

S5.2 Torsion Ladder

The first-implementation torsion ladder uses committed dimensionless defaults:

- *Ignore* if $R_S < 0.05$ (numerical noise).
- *Keep as-is / monitor* if $0.05 \leq R_S < 0.30$ (tolerable curvature).
- *Geometric fix* if $0.30 \leq R_S < 0.60$: propose a torsion-aligned split and accept it only if the S3.4 evidence gate agrees.
- *Non-linear upgrade* if $R_S \geq 0.60$: propose a local warp. If the fitted warp fails to reduce R_S below 0.60 on the patch, fall back to a torsion-aligned split, again evidence-gated.

The thresholds (0.05, 0.30, 0.60) are operational defaults, not theoretical constants. Required logs include the distribution of R_S , fraction of simplices in each ladder band, accepted torsion-split rate, warp proposal rate, and rollback/fallback rate. Retuning is warranted only if more than 25% of accepted torsion splits immediately trigger the edge-ratio fallback or if more than 10% of mini-NSF patches fall back to geometric splits after training.

S5.3 Split Placement and Edge-Ratio Fallback

A torsion-aligned split inserts the new vertex along the dominant torsion axis, typically using the longest edge projected onto the principal direction of Ω_S . The proposal must pass the local sample/evidence budget before it is scored. After constructing the candidate children, compute the edge-length ratio

$$r_{LS}(S) = \ell_{\max}(S)/\ell_{\min}(S)$$

from cached pairwise edge norms. If any proposed child has $r_{LS} > 5$, redo the placement as a centroid star split. This is the only shape-like safeguard in the pipeline; it is local to fresh split candidates and does not require a mesh-wide shape audit.

S5.4 No Per-Simplex Shape Audit

Proteus does not compute inradius/circumradius ratios, Cayley–Menger determinants, or dihedral-angle quality scores as runtime diagnostics. The reason is structural: variance-cap equilibration drives node hit masses toward a common value in regular interiors, and Hebbian edges approximate Voronoi adjacency at the active scale. The resulting stars are approximately regular without an enforced shape floor. Persistent irregularity is expected only at boundaries and dimensionality junctions, where it is informative rather than a generic numerical failure. Boundary cases are handled by the dual-flow boundary taxonomy; junctions are handled by the stuck-junction detector and freeze rule.

S5.5 Non-Linear Warp Strategies

Warp proposals are triggered by torsion that remains high after ordinary settling or by broad curvature patterns that repeatedly reject geometric splits. At a non-root recursion level, both global warp candidates and mini-NSF patches are trained in the child PCA frame fixed by S1.4, after any accepted marginal Gaussianization in that same frame. Thus the invertible maps operate on the retained intrinsic working coordinates rather than the parent ambient coordinates, lowering warp cost without discarding information admitted by the child projection audit. Let P_κ be the fraction of simplices with $R_S \geq 0.30$.

- If $P_\kappa > 50\%$, curvature is widespread and a shallow global invertible map may be proposed.
- If $P_\kappa \leq 25\%$, curvature is patchy and compact mini-NSF patches are attached only to the hottest regions.
- If $25\% < P_\kappa \leq 50\%$, use a moderate global map and mini-NSFs on the worst roughly 10% of patches.

Patch identification runs Leiden community detection on the facet-adjacency graph of hot simplices ($R_S \geq 0.60$), with resolution auto-tuned so no patch exceeds $P_{\max} = 64$ simplices. If a connected hot component cannot be split below $P_{\max}$ by increasing Leiden resolution, recursively bisect it along the longest dual-graph geodesic; if bisection still leaves a patch above $2P_{\max}$, refuse the warp and fall back to torsion-aligned geometric refinement for that component. Each mini-NSF uses 2 rational-quadratic spline coupling layers when $n_{\text{patch}} < 5000$ routed samples and 3 layers otherwise; hidden width is 64 below 5000 samples and 128 above; use 8 bins by default and 12 only when held-out likelihood underfits while R_S remains high. Train for

$$N_{\text{steps}} = \min\left(5000, \max\left(500, \frac{5n_{\text{patch}}}{B}\right)\right), \quad B = \min\left(512, \max\left(64, \frac{n_{\text{patch}}}{20}\right)\right),$$

with early stopping after 10 validation checks without improvement. Each warp proposal is accepted only if median patch torsion falls below 0.60 or drops by at least 30%, and held-out

log likelihood improves by at least one standard error over the unwarped simplex-local density. Otherwise the warp is rolled back and the system returns to the PL complex or proposes a geometric fallback.

S5.6 Warp / Complex Composition

Let P_C denote the active PCA projection for region C (the identity at levels where no rank reduction is retained), set $\mathbf{z} = P_C \mathbf{x}$, and let T_C map projected coordinates into warped coordinates $\mathbf{y} = T_C(\mathbf{z})$ where the cluster is closer to Gaussian or PL-regular. The complex density is evaluated with respect to the active PCA coordinates:

$$p_Z(\mathbf{z}) = p_Y(T_C(\mathbf{z})) |\det J_{T_C}(\mathbf{z})|.$$

Sampling reverses the path in the active model coordinates: draw $\mathbf{y}$ from the simplex-local or Gaussian summary in warped coordinates, apply T_C^{-1} into the region PCA frame, and lift through the stored PCA composition when original-coordinate samples are requested. The ambient-coordinate object is therefore a measure on the learned support rather than a full-dimensional density unless no rank reduction was retained. Residuals used by the torsion audit are measured in the active coordinate frame; when a warp is attached, gradients and residual corrections are pulled back through the warp so the explicit complex remains the domain of mass, pressure, and simplex adjacency.

Beyond the basic pullback formula, four operational rules keep warped patches compatible with the explicit complex: boundary identity, unambiguous patch ownership, periodic re-fitting after edits, and ordered composition with any global warp.

Identity-on-boundary constraint (continuity at patch edges). A patch warp T_P is constrained to be the identity on the patch boundary ∂P :

$$T_P(\mathbf{x}) = \mathbf{x}, \quad DT_P(\mathbf{x}) = I, \quad \mathbf{x} \in \partial P.$$

This is enforced operationally by (i) including ∂P vertices as fixed pre-images in the NSF training loss with weight on the order of $10^3 \lambda_{\text{data}}$, and (ii) re-evaluating boundary vertices at training-end and projecting any residual drift back to identity (a single linear correction of the leading coupling-layer bias). Without this constraint the density is discontinuous at ∂P ; with it, $\log |\det DT_P| \rightarrow 0$ and the warped complex agrees with the unwarped complex on boundary vertices.

Containing-patch resolution. For a query point $\mathbf{x}$, locate the containing simplex S^* in $\mathbf{y}$ -space using the per-level ANN; if S^* belongs to a patch P with attached T_P , use T_P for density and sampling, otherwise use the identity. Patch membership is cached at attachment time and refreshed when patches are re-segmented. Overlapping patches are not permitted: a simplex belongs to at most one patch warp at a given level.

Warp re-fit policy under structure edits. Edits inside a warped patch are scored by F_{DM} in $\mathbf{y}$ -space and applied to the warped complex; the warp T_P is retained until either (a) $K_{\text{warp}} = 64$ accepted edits have occurred in P , at which point T_P is re-trained with the updated complex as fixed structure, or (b) more than 25% of simplices in P have residual torsion ratio above 0.60, which immediately triggers a re-fit. Patch boundaries are held fixed across re-fits so the identity-on-boundary constraint continues to hold.

Composition with a global warp. When the strategy is mixed (S5.5), a global Glow T_{glob} is composed first and patch warps act on top in the Glow output frame: $\mathbf{y}_P = T_P \circ T_{\text{glob}}(\mathbf{x})$. Density and sampling chain Jacobians,

$$\log p(\mathbf{x}) = \log p_{\mathbf{y}_P}(T_P \circ T_{\text{glob}}(\mathbf{x})) + \log |\det DT_P| + \log |\det DT_{\text{glob}}|,$$

and the patch boundary identity constraint is enforced in the Glow output frame (not in the original data frame).

S6. Dual Flow and Density Evaluation

S6.1 Online Face-Pressure Tallies

When a sample $\mathbf{x}$ is assigned to a winning simplex S , compute residuals relative to the simplex barycenter and project them onto outward facet normals. Each facet receives a nonnegative pressure increment proportional to

$$\Delta \hat{p}_f \propto \max\{0, (\mathbf{x} - \bar{\mathbf{w}}_S)^\top \mathbf{n}_f\}.$$

This fractional tallying avoids winner-take-all face votes and produces smoother boundary/flux evidence for the later conservative solve.

S6.2 Conservative Reconstruction

For each simplex S with facets $f_0, \dots, f_d$, outward unit normals $\mathbf{n}_{f_i} \in \mathbb{R}^d$, and facet areas A_{f_i} , stack the incident pressures as $\mathbf{p}_S = (p_{f_0}, \dots, p_{f_d})^\top$ and define

$$A_S \in \mathbb{R}^{d \times (d+1)}, \quad (A_S)_{\cdot, i} = A_{f_i} \mathbf{n}_{f_i}.$$

Thus

$$A_S \mathbf{p}_S = \sum_i A_{f_i} p_{f_i} \mathbf{n}_{f_i}$$

is the discrete divergence-theorem sum over facets of S . At steady state on a conservative field this sum vanishes; true exterior no-flux boundary contributions are zeroed according to S6.3.

After topology selection, solve for face pressures $\{p_f\}$ by minimizing

$$\lambda \sum_f (p_f - \hat{p}_f)^2 + \sum_S \mu_S \|A_S \mathbf{p}_S\|_2^2.$$

Use normalized data weight $\lambda_f = 1$ per observed face pressure after whitening $\hat{p}_f$ by its running empirical standard deviation. The conservation weight is applied uniformly across simplices:

$$\mu_S = \frac{0.1 \bar{\lambda}_S}{\|A_S\|_F^2 + \epsilon_A}, \quad \bar{\lambda}_S = \frac{1}{d+1} \sum_{f \prec S} \lambda_f, \quad \epsilon_A = 10^{-8}.$$

The first term anchors the solution to empirical tallies; the second penalizes local conservation violations. A count-aware variant may set $\lambda_f = 1 + n_f/(1 + \bar{n}_f)$, but the baseline remains $\lambda_f = 1$ for scale-stable early runs. The quadratic problem is solved by loopy Gaussian belief propagation on the face/factor graph, with damping when spectra are poorly conditioned. Log

$$r_{\text{data}} = \frac{\sum_f (p_f - \hat{p}_f)^2}{\sum_f \hat{p}_f^2 + \epsilon}, \quad r_{\text{cons}} = \frac{\sum_S \|A_S \mathbf{p}_S\|_2^2 / (\|A_S\|_F^2 + \epsilon_A)}{\sum_f p_f^2 + \epsilon}.$$

If $r_{\text{cons}} \gg r_{\text{data}}$ after convergence, increase the constant 0.1 up to 1.0; if r_{data} rises while r_{cons} is small, decrease it toward 0.01.

After the dual-flow solve and simplex-mass normalization, log

$$\epsilon_{\text{mass}} = \left| \sum_S m_S - 1 \right|, \quad \epsilon_{\text{flux}} = \frac{\sum_S \|A_S \mathbf{p}_S\|_2^2}{\sum_f p_f^2 + \epsilon}.$$

Require $\epsilon_{\text{mass}} \leq 10^{-6}$ after normalization and warn if $\epsilon_{\text{flux}} > 10^{-3}$. Persistent flux warnings trigger boundary reclassification or local remediation in the next gate window. The two conservation residuals serve different purposes: r_{cons} is the normalized residual monitored during loopy Gaussian BP, while ϵ_{flux} is the global post-solve health check on the reconstructed field.

Online vs. offline schedule. Empirical tallies $\hat{p}_f$ are written online during sample routing. The conservative field p_f is solved offline after the accepted topology and mass field have settled. This schedule prevents double counting: transition counts select structure, online face tallies record interpolation evidence, and the offline solve converts the accepted representation into a queryable probability field.

S6.3 Boundary-Face Taxonomy

Boundary facets are classified before the dual-flow solve.

- *True manifold boundaries* carry no exterior flux.
- *Computational boundaries* arise from truncation, finite samples, or region cuts and may connect to a weak ghost reservoir.
- *Orientation seams* occur between orientation patches and require sign-consistent stitching of shared pressures. After normal alignment, shared seam pressures obey $p_f^{(P_1)} = -p_f^{(P_2)}$.

The density evaluator distinguishes no-data from no-flux. Boundary classifications and residuals are logged so persistent errors can be traced to boundary handling rather than mistaken for interior topology failure.

S6.4 Simplex-Local Density Formula

For simplex S with vertices $v_0, \dots, v_d$ and facets $f_0, \dots, f_d$, define a vertex weight by summing incident facet pressures. With barycentric coordinates $\beta_i(\mathbf{x})$, set

$$\tilde{\rho}_S(\mathbf{x}) = \sum_{i=0}^d \beta_i(\mathbf{x}) w_{v_i}^{(S)}, \quad \bar{w}_S = \frac{1}{d+1} \sum_i w_{v_i}^{(S)}.$$

The simplex-local density is

$$p(\mathbf{x} \mid S) = \frac{m_S}{|S|_d} \frac{\tilde{\rho}_S(\mathbf{x})}{\bar{w}_S}.$$

The normalization is exact because barycentric coordinates integrate uniformly over a simplex. If $|S|_d$ is numerically near zero, apply a volume floor as an arithmetic safeguard; this is not a shape diagnostic. If $\bar{w}_S = 0$ (no incident pressure on any facet of S , e.g. a fully isolated simplex), fall back to the uniform profile $p(\mathbf{x} \mid S) = m_S/|S|_d$; this is a graceful degradation rather than an error condition.

S6.5 Variants and Debugging Modes

For debugging, replace the PL profile by the uniform density $p(\mathbf{x} \mid S) = m_S/|S|_d$. This isolates topology and mass selection from interpolation choices. Other variants may use alternative face-pressure aggregations, but the canonical profile above is preferred because it is linear in the same pressures used by the dual-flow solve.

S7. Multiscale Fuzzy Memberships

S7.1 Continuous Cluster Mode Discovery

After a region stabilizes, fit a Gaussian summary in the active warped coordinate frame. With node positions $\mathbf{y}_i = T_C(\mathbf{w}_i)$ and hit weights h_i , define

$$\boldsymbol{\mu}_C = \frac{\sum_{i \in C} h_i \mathbf{y}_i}{\sum_{i \in C} h_i}.$$

The covariance combines between-node spread and within-node residual variance via the law of total covariance:

$$\Sigma_C = \frac{\sum_{i \in C} h_i [(\mathbf{y}_i - \boldsymbol{\mu}_C)(\mathbf{y}_i - \boldsymbol{\mu}_C)^\top + \mathbf{s}_i - \mathbf{m}_i \mathbf{m}_i^\top]}{\sum_{i \in C} h_i}.$$

This continuous mean is the Gaussian mode; it is not required to coincide with an AP exemplar or any individual node. The AP exemplar from S2.6 is a discrete graph representative chosen from smoothed transition similarity, not the membership mode.

S7.2 Canonical Closed-Form Membership

The canonical membership score is the Gaussian relative-density

$$\mu_C(\mathbf{x}) = \exp\left(-\frac{1}{2}(T_C(\mathbf{x}) - \boldsymbol{\mu}_C)^\top \Sigma_C^{-1}(T_C(\mathbf{x}) - \boldsymbol{\mu}_C)\right) \in (0, 1].$$

Its negative log is the Mahalanobis distance to the cluster mode in the warped frame. The score is affine-invariant within that frame because linear reparameterizations are absorbed by Σ_C . If no warp is attached, T_C is the identity and the same definition is interpreted in the local data coordinates.

S7.3 PL Reference and Diagnostic Membership

The PL reference membership is derived from the simplex-local density of S6.4:

$$\mu_C^{\text{PL}}(\mathbf{x}) = \frac{p^{(C)}(\mathbf{x})}{\max_{\mathbf{z} \in \mathcal{K}_C} p^{(C)}(\mathbf{z})}.$$

Since $\tilde{\rho}_S$ is linear in barycentric coordinates within each simplex, the maximum of the PL profile over $\mathcal{K}_C$ is attained at a vertex of some simplex. This path uses simplex masses, volumes, face pressures, and barycentric coordinates. It is the exact reference for high-fidelity density semantics and the diagnostic target for assessing whether the Gaussian summary remains adequate.

S7.4 Membership Trajectory

For a query path $C_0 \rightarrow \dots \rightarrow C_L$,

$$\boldsymbol{\mu}(\mathbf{x}) = ((C_0, \mu_{C_0}(\mathbf{x})), \dots, (C_L, \mu_{C_L}(\mathbf{x}))).$$

The cluster identities encode exact path membership; the scalar scores encode continuous closeness to each region's mode. Two points may share a prefix in the identity path while having different scores at every shared level. This is the fuzzy shared-prefix property.

S7.5 Operational Role of BMU Locator and Barycentric Coordinates

The BMU simplex $S^{(C)}(\mathbf{x})$ is transient but unavoidable for routing, recursion descent, density evaluation, and region queries. Recursion descent uses the AP cluster from S2.6 that owns the located BMU. Barycentric coordinates $\beta^{(C)}(\mathbf{x})$ are required for the S6.4 PL density profile and the PL membership of S7.3, but not for the canonical Gaussian membership of S7.2. Neither the simplex locator nor barycentric coordinates are part of the portable fuzzy membership signal.

S7.6 Portable Output Artifacts

The exported model contains per-region Gaussian summaries (μ_C, Σ_C) , optional warps T_C and inverses, simplicial complexes, simplex masses, face pressures, simplex volumes, and ANN/search indices. Query-time locators are recomputed from these artifacts. The portable downstream representation is the membership trajectory plus access to density/sampling operations at chosen levels; high-fidelity density semantics are delegated to the S6.4 PL reference path.

S8. Robustness and Edge Cases

S8.1 Missing Modalities

For structured missingness, use a boolean mask over active coordinates. Distances, residuals, and moment updates are computed only on active dimensions, optionally normalized by active count so partial observations have comparable influence. Missing zeros are treated as no observation, not as small numerical values.

Outlier handling. For heavy-tailed data, optionally apply a Huber-style robust weighting to residuals during moment updates so that a single extreme observation cannot dominate the EWMA, or winsorize the top 0.5% of residual norms before they enter $(\mathbf{m}_i, \mathbf{s}_i)$. Per-sample fractional hit contributions can additionally be capped to a small constant per sample to bound a single outlier’s influence on the routed transition counts. These are recommended defaults rather than required; the variance-cap and evidence-gate machinery already discount outliers indirectly through the rank-weighted neighborhood and statistical significance tests.

S8.2 Small Datasets and Weak Evidence

A child region must exceed a minimum sample floor before recursion, structure selection, or warp attachment is trusted. Below this floor, the scaffold may still be logged for diagnostics, but recursion is deferred and structure edits require stronger evidence margins. This prevents the hierarchy from interpreting low-count stochastic variation as genuine scale structure.

S8.3 Boundary and Off-Manifold Queries

If a query point does not lie strictly inside a simplex, project it to the nearest simplex under barycentric projection. Optionally score an off-manifold query by

$$\log p_{\text{off}}(\mathbf{x}) = \log p(\Pi_{\mathcal{K}}(\mathbf{x})) - \lambda_{\perp} \|\mathbf{x} - \Pi_{\mathcal{K}}(\mathbf{x})\|^2,$$

where $\Pi_{\mathcal{K}}$ is the nearest-simplex projection and $\lambda_{\perp}$ is a user-chosen anomaly penalty. Boundary facets use the taxonomy of S6.3. True boundaries impose no exterior flux, computational boundaries retain ghost-reservoir behavior, and orientation seams use stitched pressures.

S8.4 Dimensionality Junction Detection and Freeze Rule

For each vertex star, compute a five-indicator junction score

$$\begin{aligned} J_i = & \mathbf{1}\{\text{Dip}(\text{cnt}[v] : v \in \text{Star}(i)) > \theta_{\text{dip}}\} \\ & + \mathbf{1}\{\text{range}(d_{\text{final},v} : v \in \mathcal{N}(i)) \geq 1\} \\ & + \mathbf{1}\{\max_j |A(i, j)| > \theta_A\} \\ & + \mathbf{1}\{\text{rejectRate}_{\text{split}}(i) > \theta_{\text{rej}}\} \\ & + \mathbf{1}\{\text{q}_{90}(R_S : S \ni i) \geq 0.30\}. \end{aligned}$$

Defaults are $\theta_{\text{dip}} = 0.05$ (Hartigan dip-test p -value), $\theta_A = 0.5$, and $\theta_{\text{rej}} = 0.8$ over the last two gate windows. If $J_i \geq 3$ for two consecutive gate windows and no accepted edit reduces it, mark i as *junction-frozen*. Cross-junction splits and merges are blocked; within-limb refinements whose affected region has constant d_{final} and does not increase J_i remain eligible for the evidence gate. The freeze is lifted if $J_i < 2$ for two gate windows after subsequent local edits.

S9. Expressivity and Correspondence Theorems

S9.1 Approximation Budget

For compactly supported densities with sufficiently regular local structure, the total approximation error can be decomposed into hierarchy error, PL mesh error, warp residual, and statistical error:

$$\varepsilon_{\text{impl}} = \varepsilon_{\text{hier}} + \varepsilon_{\text{PL}} + \varepsilon_{\text{warp}} + \varepsilon_{\text{stat}}.$$

Increasing recursion depth reduces hierarchy error until sample floors or absent scale peaks stop further decomposition. Decreasing τ^* or increasing node count reduces PL error. On a C^2 d_{intr} -manifold, a sufficient scaling is

$$N_{\text{nodes}} \gtrsim \left(\frac{C_{\text{curv}}}{\varepsilon_{\text{PL}}} \right)^{d_{\text{intr}}}, \quad \tau^* \lesssim \frac{\varepsilon_{\text{PL}}^2}{C_{\text{curv}}^2}.$$

The statistical term is controlled by routed sample count,

$$H \gtrsim \frac{\log(M/\delta)}{\varepsilon_{\text{stat}}^2},$$

and recursion stops when child regions fall below $n_{\text{min}} = \max(10d, 1000)$ or when no robust Φ_C peak exists. The warp term is counted as reduced only when the accepted warp lowers median R_S below 0.60 or by at least 30% and improves held-out likelihood by at least one standard error.

S9.2 Operational Error Ledger

The implementation tracks an observable ledger:

$$\hat{E}_{\text{tot}} = w_{\text{mass}} \text{CV}_S + w_{\text{junc}}(\eta_s + \beta_{\text{max}}) + w_{\text{tors}} q_{95}(R_S) + w_{\text{ll}} \max(0, -\Delta \ell_{\text{heldout}}) + w_{\text{stat}} \Delta_{\text{stat}}.$$

A level is expressively saturated when mass CV, torsion, held-out improvement, and statistical scale are below thresholds for consecutive gate windows, or when the remaining residual is classified as a junction/boundary residual by S8.4.

S9.3 Simplex Equilibrium Implies Effective Node Equilibrium

Let V_i be the Voronoi-like cell of node i and let C range over M top-dimensional simplices on a complex with N nodes. Define the barycentric Voronoi share

$$\alpha_{i,C} := P(V_i \cap C)/P(C), \quad \sum_{i \in V(C)} \alpha_{i,C} = 1.$$

The deterministic decomposition is

$$P(V_i) = \sum_{C \ni i} \alpha_{i,C} P(C),$$

where $\alpha_{i,C}$ is the share of simplex C associated with vertex i . Let $s_i = |\{C : i \in C\}|$, let $\bar{s} = (d+1)M/N$, and assume $|s_i/\bar{s} - 1| \leq \eta_s$. Let

$$\Delta_M(H, \delta) = \sqrt{\frac{\log(2M/\delta)}{2H}}, \quad \Delta_N(H, \delta) = \sqrt{\frac{\log(2N/\delta)}{2H}},$$

and define $\bar{\alpha}_d = 1/(d+1) + \beta_{\max}$. If empirical simplex masses satisfy

$$|\hat{P}(C) - 1/M| \leq \epsilon_{\text{sim}}/M \quad \text{for all } C,$$

then, on the joint multinomial concentration event for simplex and Voronoi counts, for every node i ,

$$\left| \hat{P}(V_i) - \frac{1}{N} \right| \leq \underbrace{\frac{\eta_s + (1 + \eta_s)(d+1)\beta_{\max}}{N}}_{\text{structural}} + \underbrace{\bar{\alpha}_d s_i \frac{\epsilon_{\text{sim}}}{M}}_{\text{equilibrium}} + \underbrace{\bar{\alpha}_d s_i \Delta_M(H, \delta) + \Delta_N(H, \delta)}_{\text{sampling}}.$$

The three terms separate persistent junction/boundary residual, residual algorithmic equilibration tolerance, and finite-sample fluctuation.

S9.4 Steady-State Star Regularity

The theory uses no enforced shape-quality floor. Define

$$\beta_{\max} := \max_{(i,C): i \in V(C)} \left| \alpha_{i,C} - \frac{1}{d+1} \right|.$$

In a regular-interior region with uniform intrinsic dimension and support bounded away from boundaries, variance-cap equilibration drives node hit masses toward a common value and Hebbian edges approximate Voronoi adjacency. In that limit $\beta_{\max} \rightarrow 0$ and $\eta_s \rightarrow 0$. Persistent η_s or $\beta_{\max}$ indicates a boundary or dimensionality junction and is handled operationally by S8.4 and the S6.3 boundary taxonomy rather than forced away by mesh-quality audits.

S10. Generative Model and Identifiability

S10.1 Mass Field and Geometry-Derived Router

For a fixed complex, let $m = \{m_S\}$ be a simplex mass field. Geometry induces conductances κ_{iS} from node i to incident simplex S and κ_{ijS} from node i through simplex S toward neighbor j . The geometry-derived router is

$$q(j \mid i; m) = \frac{\sum_{S \ni i, j} \kappa_{ijS} m_S}{\sum_{S \ni i} \kappa_{iS} m_S}.$$

This router connects the explicit complex to observed transition statistics and to the generative interpretation of simplex masses.

The committed canonical conductance uses the opposite-facet area. For a d -simplex S with vertex $i \in S$, let A_{iS} be the $(d-1)$ -volume of the facet opposite i . Define

$$\kappa_{iS} = A_{iS}, \quad \kappa_{ijS} = \frac{A_{iS}}{d} \quad \text{for } j \in S, j \neq i.$$

This is the uniform facet-share rule, and it satisfies

$$\sum_{j \in S, j \neq i} \kappa_{ijS} = \kappa_{iS}.$$

It is positive on nondegenerate complexes, scale-covariant in the mesh geometry, and cheap to compute.

S10.2 Relation to the Runtime DM Gate

Treat outgoing counts at node i as multinomial draws given the geometry-derived router:

$$\{n_{i \rightarrow j}\}_j \mid n_i, m \sim \text{Multinomial}(n_i, q(\cdot \mid i; m)).$$

The likelihood factor depending on m is

$$\phi_i(m) \propto \prod_{j \in \mathcal{N}(i)} q(j \mid i; m)^{n_{i \rightarrow j}}.$$

The mass field m is therefore the generative latent in the model family, but the runtime structure-selection gate does not estimate m by iterative MAP for every candidate. Instead, the gate scores outgoing transition counts directly with the BDeu/Dirichlet–multinomial marginal of S3.4, which integrates the node-local categorical probabilities analytically. This makes candidate comparison closed-form and prevents iteration-budget artifacts. Identifiability of m remains important because it justifies that accepted transition structure corresponds to a well-defined mass field on the complex.

MaxEnt/MAP connection. With a flat prior on m , maximizing the joint log-likelihood gives the MaxEnt/MLE estimator

$$\hat{m} = \arg \max_{m \in \Delta} \sum_i \sum_j n_{i \rightarrow j} \log q(j \mid i; m), \quad \Delta = \left\{ m : m_S \geq 0, \sum_S m_S = 1 \right\}.$$

Adding $\log \psi(m)$ for any prior ψ from S10.3 yields the MAP estimator under that prior. The runtime F_{DM} gate of S3.4 corresponds to the closed-form Bayesian-evidence limit of this same model under the BDeu prior, with m integrated out analytically; the MaxEnt/MAP form is therefore the deterministic analogue of the runtime gate, useful as a cross-check when m is needed explicitly (e.g. for sampling).

S10.3 Priors and BNP Correspondence

Natural priors over m include:

- *Simplex-wide Dirichlet*: $m \sim \text{Dirichlet}((\beta_S)_{S \in \mathcal{S}})$.
- *Logistic-Gaussian MRF*: $\theta \sim \mathcal{N}(0, \tau_L^{-1} L^+)$ on the dual-graph Laplacian and $m_S = \exp(\theta_S) / \sum_{S'} \exp(\theta_{S'})$.
- *Hierarchical Dirichlet / Pólya tree*: recursively split simplex sets and place Dirichlet priors at each branch.
- *Normalized Gamma random measure*: draw $\tilde{m}_S \sim \text{Gamma}(a_S, 1)$ and set $m_S = \tilde{m}_S / \sum_{S'} \tilde{m}_{S'}$.

The AP/BDeu machinery provides the finite-dimensional counterpart of BNP clustering behavior while remaining tied to explicit mesh geometry.

Cross-scale (multiscale) Dirichlet refinement. For a coarse cell $P \in \mathcal{S}^{(s)}$ with children $\text{ch}(P) \subset \mathcal{S}^{(s+1)}$ at the next recursion level, the refinement step can be cast as a Dirichlet-tree update

$$m_{\text{ch}(P)}^{(s+1)} \mid m_P^{(s)} \sim \text{Dir}(\alpha_s \pi_{\text{ch}(P)}^{(s)} m_P^{(s)}), \quad \sum_{C \in \text{ch}(P)} m_C^{(s+1)} = m_P^{(s)}.$$

This is the multiscale Dirichlet refinement used by Pólya trees, applied here to a simplicial mesh. Proteus’ warm-start procedure (where the equilibrated state at scale s initializes the search at $s+1$) is the deterministic analogue of using the posterior at one level as the prior for the next, and the variance-cap split rule plays the role of a data-driven Bayes-factor test for cell refinement.

S10.4 Star Matrix Identifiability

Identifiability is checked locally by star matrices that map simplex masses in $\text{Star}(i)$ to transition probabilities from node i . If each star has full rank modulo the one-dimensional scaling direction and the dual graph is connected, the full set of routers identifies m up to the simplex constraint $\sum_S m_S = 1$. At runtime, candidate edits perform a dry-run conditioning check using $\sigma_{\min}/\sigma_{\max}$ on affected stars. Edits that fail the check cannot claim an F_{DM} improvement; they are rejected or treated as geometry-only remediation with likelihood terms quarantined.

The remainder of this section makes that statement operational: Fisher information explains why rank matters, the runtime threshold defines the numerical test, and the dynamic preservation rule states when an edit is allowed to use likelihood evidence.

Fisher-information identity. Up to normalization, the star matrix K_i is the Jacobian of $q(\cdot \mid i; m)$ with respect to $m^{(i)} = m|_{\text{Star}(i)}$ at the canonical κ . The local Fisher information of the per-node transition likelihood is therefore

$$I_i(m) = n_i K_i^\top \text{diag}(1/q(\cdot \mid i; m)) K_i.$$

Local rank condition (A1) is exactly nonsingularity of $I_i(m)$ on the tangent of the simplex at m ; combined with dual-graph connectivity (A2) this yields global identifiability and standard MLE consistency as $\min_i n_i \rightarrow \infty$.

Runtime conditioning threshold. Compute $\rho_i^* = \sigma_{\min}(K_i)/\sigma_{\max}(K_i)$ at every star whose topology has just changed; default flag threshold $\rho_{\min} = 10^{-4}$ (with conservative 10^{-3} for noise-sensitive runs). Stars with $\rho_i^* < \rho_{\min}$ are marked *router-ill-conditioned*.

Dynamic preservation rule. For every proposed edit, compute ρ_i^* on the affected stars both before and after applying the candidate topology in a dry run. An edit is *evidence-bearing* only if every affected post-edit star satisfies $\rho_i^* \geq \rho_{\min}$ and the affected dual subgraph remains connected. If this condition fails, the edit cannot claim an F_{DM} improvement; it may still be accepted as *geometry-only remediation* when it improves an explicit shape or boundary diagnostic that caused the ill-conditioning, but in that case all likelihood terms from the affected ill-conditioned stars are quarantined until conditioning is restored.

Trajectory-level conservative identifiability. Along any trajectory obeying the dynamic preservation rule, every accepted edit that uses transition-count evidence does so only in locally identifiable stars. Geometry-only remediation edits may pass through ill-conditioned states, but those states contribute no likelihood terms and therefore cannot create spurious evidence improvements. Consequently, the mass field used by the evidence gate is identifiable at every evidence-bearing decision point, and ill-conditioned transients remain conservative numerical states rather than alternative statistically valid mass explanations.

S11. Stage 1 / SVD Limit

S11.1 Setting and Kernel Mapping

Consider a local Gaussian window centered at node position $\mathbf{w}_i$ with bandwidth calibrated by

$$\sigma = \frac{\sqrt{\tau}}{c_{d,k}},$$

where $c_{d,k}$ maps the variance cap to the expected k -NN radius under local isotropy. The EWMA residual moments estimate a kernel-weighted local mean and covariance. Oja directions track the dominant eigenvectors of this local covariance.

S11.2 Conditions for Equivalence

The reduction to Gaussian-weighted PCA/SVD holds under the following simultaneous limits/conditions:

1. topology and node positions are fixed over the moment-estimation window;
2. learning rate is small enough that EWMA moments track stationary kernel expectations;
3. ANN recall is high, so rank-weighted neighborhoods approximate true local neighborhoods;
4. sampling is locally stationary and smooth at bandwidth σ ;
5. no split/prune/edit event occurs during the comparison interval;
6. the rank-weighted neighborhood is isotropic up to the error terms in S12.1.

Under these conditions, moment estimates converge to kernel-weighted covariance estimates and Oja updates recover local principal directions.

S11.3 Consequences and Non-Equivalences

The equivalence explains why Stage 1 recovers intrinsic variance directions and why the variance cap behaves like a local resolution criterion. It does not reduce the full algorithm to SVD: adaptive node placement, Hebbian topology, evidence-gated splits/prunes, AP recursion, and Stage 2 generative lifting are genuinely algorithmic additions.

S12. Dynamic Convergence

S12.1 Lyapunov Candidate and Standing Hypotheses

The dynamic analysis uses a joint potential

$$L(t) = L_{\text{vertex}}(t) + F_{\text{DM}}(M(t); \text{counts}(t)), \quad L_{\text{vertex}}(y) = \sum_i \int_{V_i(y)} \|x - y_i\|^2 p(x) dx.$$

Deferred nudges are handled by virtual positions $\mathbf{y}_i = \mathbf{w}_i + \mathbf{a}_i$. Once deferred moves are processed, $\|\mathbf{a}_i\| < \delta_{\min}$, so reported positions inherit the virtual-position bound by a local Taylor comparison:

$$|L_{\text{vertex}}(w) - L_{\text{vertex}}(y)| \leq C_{\text{def}} \delta_{\min} + \frac{1}{2} N L_{\nabla} \delta_{\min}^2, \quad C_{\text{def}} \leq 2DN$$

on compact separated codebooks.

On the separated set $\mathcal{K}_\Delta = \{y : \min_{i \neq j} \|y_i - y_j\| \geq \Delta\}$, with bounded hyperplane marginals near Voronoi bisectors, L_{vertex} is C^1 and

$$\nabla_{y_i} L_{\text{vertex}}(y) = -2P(V_i)(\bar{x}_i - y_i), \quad \|\nabla_{y_i} L_{\text{vertex}}\| \leq 2D.$$

Set $\alpha = \ln 2/k$. For rank weight $\omega_i(X; y)$, define

$$\sigma_{i,\text{rw}}^2(y) = \frac{\mathbb{E}[\omega_i(X; y) \|X - y_i\|^2]}{\mathbb{E}[\omega_i(X; y)]}.$$

The clean tier assumes fixed topology, frozen counts, active-node recurrence $\mathbb{E}[\omega_i(X; y)] \geq \mu_{\min} > 0$, compact separated codebooks, bounded innovations, and strict cap margin

$$\sigma_{i,\text{rw}}^2(y) \leq (1 - \zeta)\tau_i, \quad \zeta > 0.$$

The rank-weighted drift differs from Voronoi-gradient drift by a bounded perturbation $\varepsilon^{\text{rank}}$. Under the S11 isotropic-neighborhood assumptions, one sufficient scaling is

$$\|\varepsilon_i^{\text{rank}}\| \leq \frac{\alpha}{2} \left(1 - \frac{\sigma_i^2}{\tau_i}\right) M_i^{\text{nb}} \sqrt{\tau_i} \delta_{\text{SVD},i},$$

with

$$\delta_{\text{SVD},i} \leq C_d/k_{\text{NN}} + \varepsilon_{\text{ANN}} + C_\partial \text{vol}(\partial)/\text{vol}(B_\sigma) + C_a \|\nabla \log p\| \sqrt{\tau_i}.$$

The fixed- k algorithm targets a small invariant neighborhood rather than point convergence.

S12.2 Lemma 1A: Fixed Topology and Frozen Counts

Statement. Fix topology M , freeze evidence counts, and consider a no-edit interval on which the standing hypotheses of S12.1 hold. Let β_0 be the BMU rank weight and define

$$\bar{\Gamma}_i(t) = \frac{\beta_0}{4} \left(1 - \frac{\sigma_i^2(t)}{\tau_i}\right), \quad c_t = \alpha \sum_i \bar{\Gamma}_i(t) \|\nabla_{y_i} L_{\text{vertex}}(y(t))\|^2.$$

Then

$$\mathbb{E}[L_{\text{vertex}}(y(t+1)) \mid \mathcal{F}_t] \leq L_{\text{vertex}}(y(t)) - c_t + C_\alpha \alpha^2 + C_{\text{rank}} \alpha \|\varepsilon^{\text{rank}}\|^2.$$

Consequently, in any local basin where

$$\sum_i \bar{\Gamma}_i \|\nabla_{y_i} L_{\text{vertex}}\|^2 \geq \mu(L_{\text{vertex}}(y) - L_*^M),$$

the fixed- k dynamics enter the neighborhood

$$\limsup_{t \rightarrow \infty} \mathbb{E}[L_{\text{vertex}}(y(t)) - L_*^M] \leq C_{\text{loc}}(C_\alpha \alpha + C_{\text{rank}} \|\varepsilon^{\text{rank}}\|^2),$$

with $O(\sqrt{\alpha})$ stationary state fluctuations. Reported positions w add the deferred-nudge tolerance of S12.1.

Proof sketch. The EWMA variance recursion contracts to $\sigma_{i,\text{rw}}^2(y)$ at effective rate at least $\alpha\mu_{\min}$ with stationary mean-square error $O(\alpha)$. The expected virtual-position drift on cap-satisfied intervals is

$$\mathbb{E}[\Delta y_i \mid \mathcal{F}_t] = -\alpha \bar{\Gamma}_i(t) \nabla_{y_i} L_{\text{vertex}}(y(t)) + \alpha \varepsilon_i^{\text{rank}} + O(\alpha^{3/2}).$$

A Taylor expansion on $\mathcal{K}_\Delta$ gives a negative gradient term plus $O(\alpha^2)$ curvature error and the rank-bias inner product. Young's inequality absorbs the rank term into $C_{\text{rank}} \alpha \|\varepsilon^{\text{rank}}\|^2$ and part of the negative drift. Standard constant-gain stochastic-approximation results then give the displayed invariant neighborhood.

S12.3 Lemma 1B: Streaming and Topology Edits

Standing assumptions. The streaming/edit tier assumes: (A1) post-edit codebooks remain compact and separated; (A2) active nodes retain visitation at least $\mu_{\min}$; (A3) continuous-gradient intervals satisfy the strict cap margin of S12.1, while over-cap nodes are treated as split-trigger regimes; and (A4) sample innovations are bounded or sufficiently mixing.

SI-level facts. The remaining edit-tier conditions are discharged by their owning SI sections: (A5) S3.3 gives a uniform per-edit positive jump bound B_p for guarded prunes/merges; (A6) S3.6 gives finite-epoch positive-jump budget $A_T < \infty$ and an indefinite-streaming diagnostic; (A7) S12.1 gives explicit rank-bias scaling; and (A8) S3.6 hysteresis plus post-edit settle windows, together with S12.6, prevent edit transients from accumulating at linear rate.

Statement. Let

$$A_T = \sum_{t < T} [\Delta L_{\text{vertex}}^{\text{prune/merge}}(t)]_+.$$

Under (A1)–(A4) and the SI-level facts above,

$$\frac{1}{T} \sum_{t < T} \mathbb{E}[c_t] \leq \frac{L_{\text{vertex}}(y(0)) - L_{\text{inf}}}{T} + C_\alpha \alpha^2 + C_{\text{rank}} \alpha \|\varepsilon^{\text{rank}}\|^2 + \frac{\mathbb{E}A_T}{T} + o(1).$$

After normalizing by the constant-gain drift scale, topology edits add the tolerance

$$\limsup_T \frac{\mathbb{E}A_T}{\alpha T}.$$

This term vanishes in finite-epoch operation with finite total jump budget, and is a monitored engineering quantity in indefinite streaming.

Proof sketch. Between accepted edits, Lemma 1A applies on each cap-satisfied fixed-topology interval. Splits do not increase nearest-center distortion pointwise; prunes and merges can increase it, but only by the guarded local bound accumulated in A_T . Over-cap nodes are charged to split-trigger regimes rather than the continuous-gradient proof. Summing Lemma 1A over fixed-topology intervals and adding jump/transient budgets yields the inequality.

S12.4 Edit-Decision Monotonicity

Statement. At gate-scoring instants, with counts frozen during comparison, accepted edits satisfy:

1. continuous intervals follow Lemma 1A;
2. accepted discrete edits have $\Delta F_{\text{DM}} \leq -\log \tau_{\text{BF}}$;
3. accepted splits have $\Delta L_{\text{vertex}} \leq 0$;
4. accepted prunes/merges have $0 \leq \Delta L_{\text{vertex}} \leq B_p$;
5. rejected edits change no topology at the edit instant.

Thus accepted splits decrease the frozen-count joint score. Accepted prunes/merges decrease it when $\log \tau_{\text{BF}} > B_p$, and otherwise contribute only through the explicit positive-jump budget of Lemma 1B.

Proof sketch. The evidence inequality is exactly the acceptance rule of S3.4. The split sign follows from adding a center to a nearest-center distortion objective. The prune/merge bound follows from S3.3 local guards or the bounded-support fallback. Rejected edits leave the state unchanged at the decision instant.

S12.5 Cross-Scale Inheritance

Statement. Under the explicit T1–T3 policies of S4.4, the joint potential inherits a finite value across scale and recursion boundaries with bounded initialization error.

Proof sketch.

- *T1 cross-grid:* positions, links, hit counts, and transition counts carry forward unchanged; moment shrink and nudge reset affect subsequent dynamics but not the instantaneous value of L_{vertex} or F_{DM} .
- *T2 parent-to-child:* positions and intrinsic dimensions carry forward, but counts/moments reset. The child L_{vertex} is computed on the restricted data marginal and is bounded on compact support; F_{DM} returns to a finite prior-count value bounded by the child node count and maximum branching factor.
- *T3 Stage 1 to Stage 2:* positions, counts, and link structure carry forward. Moment shrink affects only subsequent dynamics. Intrinsic-dimension updates shift BDeu prior factors by $O(\log d)$ per affected node.

In all cases, the next trajectory resumes Lemma 1A drift from a bounded initialization error.

S12.6 EWMA Noise Control

Statement. After an accepted split at time t_0 , the transient perturbation to neighboring variance estimates satisfies, with probability at least $1 - \delta$,

$$|\sigma_j^2(t) - \sigma_{j,*}^2| \leq C_R e^{-(t-t_0) \ln 2/k} + R^2 \sqrt{2\alpha \log(2/\delta)},$$

where R is the compact-support residual bound and $C_R = O(R^2)$.

Proof sketch. Write the post-edit EWMA recursion as

$$\sigma_j^2(t+1) = (1 - \alpha)\sigma_j^2(t) + \alpha r_t^2, \quad r_t^2 = \sigma_{j,*}^2 + \xi_t,$$

with $\mathbb{E}[\xi_t] = 0$ and $|\xi_t| \leq R^2$. Then

$$\sigma_j^2(t) - \sigma_{j,*}^2 = (1 - \alpha)^{t-t_0}(\sigma_j^2(t_0) - \sigma_{j,*}^2) + \alpha \sum_{s=t_0+1}^t (1 - \alpha)^{t-s} \xi_s.$$

The deterministic term decays with half-life k . Azuma–Hoeffding gives the displayed stochastic radius. Lemma 1B uses the condition that these finite transients do not accumulate at linear rate across edits. When partition-aligned shadow moments are maintained (S2.3.2), the splitting node and its child enter the post-split trajectory at the EWMA precision implied by their inherited $h_i^\pm \leq h_i$, so the deterministic transient $|\sigma_j^2(t_0) - \sigma_{j,*}^2|$ is bounded by the shadow variance rather than the parent’s over-cap aggregate; the displayed bound remains valid with C_R replaced by the shadow-local residual bound.

S12.7 Status Summary and Remaining Assumptions

- **Lemma 1A (proved):** fixed topology, frozen counts, virtual positions, strict cap margin, active-node recurrence, and bounded rank bias imply a fixed- k constant-gain neighborhood theorem for L_{vertex} .
- **Lemma 1B (conditional):** streaming and topology edits preserve the same neighborhood under standard SA assumptions (A1)–(A4), with S3.3/S3.6/S12.6 discharging the edit-budget and transient facts for finite-epoch operation.

- **Lemmas 2–4:** edit-decision monotonicity, cross-scale inheritance, and EWMA noise control supply the supporting pieces needed by Lemma 1B.
- **Operational tier:** junction freezing and expressivity stopping are implementation-ready diagnostics. Future work may sharpen canonical junction lower bounds and approximation constants.

S13. Reference Algorithms

S13.1 Stage 1 at Fixed Scale

Algorithm 1 Stage 1 at fixed τ

```

1: procedure RUNTOEQUILIBRIUM( $X, \tau$ )
2:   Initialize or warm-start scaffold  $G_C = (\mathcal{V}_C, \mathcal{E}_C)$ 
3:   repeat
4:     Sample  $\mathbf{x}$  from  $X_C$ 
5:      $\mathcal{N}_k \leftarrow \text{QUERYANN}(\mathbf{x}, k)$ 
6:      $\text{APPLYRANKWEIGHTEDUPDATES}(\mathbf{x}, G_C, \mathcal{N}_k)$ 
7:     for each updated node  $i$  do
8:       if  $\|\mathbf{a}_i\| \geq \delta_{\min}$  then
9:          $\mathbf{w}_i \leftarrow \mathbf{w}_i + \mathbf{a}_i$ ;  $\mathbf{a}_i \leftarrow 0$ ;  $\text{REFRESHANN}(i)$ 
10:      end if
11:      if  $\sigma_i^2 > \tau_{\text{local},i}$  then
12:        Queue variance-cap split proposal for  $i$ 
13:      end if
14:    end for
15:    Apply link/node pruning gauntlets from S3.1–S3.3
16:  until fixed-scale stabilization diagnostics pass
17:  return stabilized scaffold  $G_C$  and response statistics
18: end procedure

```

S13.2 Rank-Weighted Neighbor Updates

Algorithm 2 Rank-weighted neighbor updates

```

1: procedure APPLYRANKWEIGHTEDUPDATES( $\mathbf{x}, G_C, \mathcal{N}_k$ )
2:   for  $r = 1, \dots, k$  in rank order do
3:      $i \leftarrow \mathcal{N}_k[r]$ ;  $\omega_r \leftarrow 2^{-(r-1)}$ 
4:      $\mathbf{e}_i \leftarrow \mathbf{x} - \mathbf{w}_i$ 
5:      $\mathbf{m}_i \leftarrow (1 - \alpha\omega_r)\mathbf{m}_i + \alpha\omega_r\mathbf{e}_i$ 
6:      $\mathbf{s}_i \leftarrow (1 - \alpha\omega_r)\mathbf{s}_i + \alpha\omega_r(\mathbf{e}_i^{\circ 2})$ 
7:      $\rho_i \leftarrow \|\mathbf{m}_i\|/(\sigma_i + \varepsilon)$ 
8:      $\mathbf{a}_i \leftarrow \mathbf{a}_i + \eta_{\text{cent}}\rho_i\mathbf{m}_i$ 
9:      $\text{UPDATEOJA}(\mathbf{u}_i, \mathbf{e}_i)$ 
10:     $h_i \leftarrow h_i + \omega_r$ 
11:   end for
12:   Update directed transition counters for first/second nearest co-activation
13: end procedure

```

S13.3 Stage 2 Simplex-Native Updates

Algorithm 3 Simplex-native updates (Stage 2)

```

1: procedure APPLYSIMPLEXNATIVEUPDATES( $\mathbf{x}, \mathcal{K}_C$ )
2:    $S \leftarrow \text{LOCATEWINNINGSIMPLEX}(\mathbf{x}, \mathcal{K}_C)$ 
3:   Order vertices  $v \in V(S)$  by increasing  $\|\mathbf{x} - \mathbf{w}_v\|$ 
4:   for each vertex  $v$  in rank order do
5:     assign rank weight  $\omega_r = 2^{-(r-1)}$ 
6:      $\mathbf{e}_v \leftarrow \mathbf{x} - \mathbf{w}_v$ 
7:      $\mathbf{m}_v \leftarrow (1 - \alpha\omega_r)\mathbf{m}_v + \alpha\omega_r\mathbf{e}_v$ 
8:      $\mathbf{s}_v \leftarrow (1 - \alpha\omega_r)\mathbf{s}_v + \alpha\omega_r(\mathbf{e}_v^{\circ 2})$ 
9:      $\rho_v \leftarrow \|\mathbf{m}_v\|/(\sigma_v + \varepsilon)$ 
10:     $\mathbf{a}_v \leftarrow \mathbf{a}_v + \eta_{\text{cent}}\rho_v\mathbf{m}_v$ 
11:   end for
12:   Accumulate empirical facet pressures  $\{\hat{p}_f : f \prec S\}$  using S6.1
13:   Update torsion statistics for  $S$  and queue proposals whose diagnostics cross thresholds
14:   do not accept topology edits here; queued edits are processed by the S3.6 gate
15: end procedure

```

S13.4 Greedy Chaining

S14. Evaluation Protocols and Defaults

S14.1 Datasets and Splits

Synthetic benchmarks should include known topological and geometric ground truth: nested manifolds, variable-density curves/surfaces, linked components, dimensionality junctions, and high-curvature patches. Real benchmarks should be chosen for local structure and query relevance rather than only classification accuracy. Use fixed train/test splits or repeated random splits with logged seeds.

S14.2 Metrics

Metrics include persistent-homology distances or topology summaries where ground truth exists, reconstruction/projection error, held-out log-likelihood under the reconstructed density, sample quality (e.g. MMD or nearest-neighbor two-sample tests), membership-trajectory stability, accepted-edit counts, and diagnostic residuals. Each metric should map to a named architectural claim.

Persistent homology. Use Vietoris–Rips complexes up to maximum dimension 2, with filtration up to $1.5\sigma_*$ where $\sigma_* = \sqrt{\tau^*}$ is the active scale. Report bottleneck and 2-Wasserstein distances between the learned complex and the ground-truth complex per dimension. Average across at least 5 random seeds.

Maximum mean discrepancy. Use an RBF kernel with bandwidth set by the median heuristic on the combined sample. Default to 1,000 samples drawn from the model and 1,000 samples from the held-out test set, repeated over 5 seeds; report mean $\pm$ std and a permutation test p -value.

Held-out log-likelihood. Evaluate $\log p^{(C)}(\mathbf{x})$ on the held-out split using the simplex-local density of S6.4 at the user’s chosen recursion level. The default level is the leaf containing the BMU of $\mathbf{x}$; report results at the leaf level and at one or two intermediate levels to expose multiscale behavior.

Default Stage 1/Stage 2 settings. Stage 1 uses $k = 8$ and $\alpha = \ln 2/8$ on a geometric scale grid with ratio $r = 1/\sqrt{2}$. Stage 2 uses $k \approx d$ and $\alpha = \ln 2/k$. The Stage 1 stabilization criterion is $\text{CV}(\sigma_i^2) < \beta_{\text{cv}} \sqrt{2/k}$ over mature nodes (those with enough BMU events for their hard-Voronoi variance estimate to be reliable), with default buffer $\beta_{\text{cv}} = 1.5$. The threshold is derived from the EWMA noise floor: a half-life- k EWMA has effective sample size approximately k , so the variance estimator’s standard relative error is $\sqrt{2/k}$ per node; the buffer accounts for residual spatial heterogeneity. For $k = 8$ the default threshold is ≈ 0.75 . The neighbor-normalized incoherence $\text{CV}(\tilde{\rho})$ remains a useful diagnostic for spatial and bias heterogeneity but is not used as the stopping criterion because the incoherence ratio combines hard-Voronoi σ_i with soft-routing $\mathbf{m}_i$ (S2.3.1) and is fragile to neonatal-node transients where $\sigma_i \rightarrow 0$. The torsion ladder admits regions when the 95-th percentile R_S falls below 0.30. Standard ablations are: remove the Stage 1 scaffold (force a fixed scale), disable the torsion audit, replace the dual-flow reconstruction with uniform simplex density, and disable the evidence gate.

S14.3 First-Implementation Defaults

Defaults are listed here for implementers while remaining defined in their owning sections.

Object	Default	Owner	Status
α	$\ln 2/k$	S2.1	theory/operational
η_{cent}	$\kappa(1-r)/k, \kappa = 0.5$	S2.2	operational
$s_{\text{control}} \mapsto \tau_i$	$-d_i \log(1 - s_{\text{control}})$	S2.4	theory
scale grid	$r = 1/\sqrt{2}$	S2.5	operational
Φ_C peak floor	5% of grid max	S2.5.1	empirical
$\alpha_{0,i}$	$1/(d_{\text{final},i} + 1)$	S2.7	theory
z_{Wilson}	1.2816	S3.1	statistical
node prune hit	$h_i < 0.5 \bar{h}_{\mathcal{N}(i)}$	S3.2	operational
node prune variance	$\sigma_i^2 < 0.5 \tau_i$	S3.2	operational
ρ_{merge}	$\leq 2C_Q(d)$	S3.3	operational
p_{merge}	$\leq \bar{h}/N$	S3.3	operational
v_{merge}	≤ 1	S3.3	operational
$\log \tau_{\text{BF}}$	[1, 3] for splits	S3.4	empirical
W	$\max(N_{\text{nodes}}^{(C)}, 4k \mathcal{Q}_C)$	S3.6	operational
T_{hyst}	$2W$	S3.6	operational
edit budget	$N_{\text{nodes}}^{(C)} / \log N_{\text{nodes}}^{(C)}$	S3.6	operational
γ_0	0.7	S4.4	empirical
R_S thresholds	0.05, 0.30, 0.60	S5.2	empirical
r_{LS}	$\ell_{\text{max}}/\ell_{\text{min}} > 5$	S5.3	operational
P_κ	25%, 50%	S5.5	empirical
P_{max}	64 simplices	S5.5	operational
λ_f	1	S6.2	operational
μ_S	$0.1\bar{\lambda}_S/(\ A_S\ _F^2 + \epsilon_A)$	S6.2	operational
ϵ_A	10^{-8}	S6.2	numerical
ϵ_V	10^{-3}	S6.4	numerical
n_{min}	$\max(10d, 1000)$	S8.2	operational
θ_{dip}	0.05	S8.4	empirical
θ_A	0.5	S8.4	empirical
θ_{rej}	0.8	S8.4	empirical

Defaults marked empirical are expected to be retuned after benchmark evidence; defaults marked theory/statistical follow directly from the corresponding derivation or test convention.

S14.4 Logging Checklist

Required logs are per region C unless otherwise stated.

- *Scale discovery:* $\Phi_C(\tau)$, $V_C(\tau)$, grid points evaluated, peak brackets, optimizer refinements, selected τ_C^* , and stabilization failures.
- *Scaffold statistics:* hit CV, variance-cap violations, deferred-nudge rate, ANN recall diagnostics, intrinsic-dimension estimates, and AP cluster sizes.
- *Evidence gate:* proposal queue entries, affected regions, rerouted counts, F_{DM} keep/edit scores, Bayes-factor margins, stale-proposal invalidations, accepted/rejected decisions, hysteresis blocks, and edit-budget use.
- *Torsion and warps:* R_S distributions, ladder band fractions, edge-ratio fallback counts, P_κ , mini-NSF patch sizes, train steps, held-out likelihood deltas, pre/post torsion quantiles, rollback counts.

- *Dual flow and density:* r_{data} , r_{cons} , ϵ_{mass} , ϵ_{flux} , boundary classifications, message convergence, and volume-floor uses.
- *Membership and stopping:* $\mu_C - \mu_C^{\text{PL}}$ residuals, junction scores J_i , freeze/lift events, mass CV, $\eta_s + \beta_{\text{max}}$, held-out likelihood improvement, and final stopping reason.

Algorithm 4 Greedy Chaining: node-seeded simplex discovery

```

1: procedure GREEDYCHAINING( $G_C, \{d_{\text{final},v}\}$ )
2:    $\mathcal{S}_C \leftarrow \emptyset$  ▷ canonical-form simplexes
3:    $\text{cnt}[v] \leftarrow 0$  for all  $v \in \mathcal{V}_C$ 
4:    $\text{queue} \leftarrow \text{PRIORITYQUEUE}(\mathcal{V}_C, \text{key} = \text{deg}, \text{descending})$ 
5:    $\text{visited} \leftarrow \emptyset$ 
6:   while  $\text{queue}$  not empty do
7:      $i \leftarrow \text{queue.pop}()$ 
8:     if  $i \in \text{visited}$  then continue
9:     end if
10:     $\text{visited} \leftarrow \text{visited} \cup \{i\}$ 
11:     $d \leftarrow d_{\text{final},i}$ 
12:    if  $|\mathcal{N}(i)| < d$  then
13:      LOGORPHAN( $i$ ); continue
14:    end if
15:    if  $\text{cnt}[i] \geq d + 1$  then continue
16:    end if
17:    for each  $d$ -subset  $T \subseteq \mathcal{N}(i)$  do
18:      if not ISCLIQUE( $T, G_C$ ) then continue
19:      end if
20:       $\sigma \leftarrow \text{CANONICALSORT}(\{i\} \cup T)$ 
21:      if  $\sigma \in \mathcal{S}_C$  then continue
22:      end if
23:       $\mathcal{S}_C \leftarrow \mathcal{S}_C \cup \{\sigma\}$ 
24:      for  $v \in \sigma$  do
25:         $\text{cnt}[v] \leftarrow \text{cnt}[v] + 1$ 
26:      end for
27:      if  $\text{cnt}[i] \geq d + 1$  then break
28:      end if
29:    end for
30:    for  $j \in \mathcal{N}(i)$  do
31:      if  $j \notin \text{visited}$  then
32:         $\text{queue.push}(j)$ 
33:      end if
34:    end for
35:  end while
36:  log cnt distribution ▷ junction signature
37:  return  $\mathcal{S}_C$ 
38: end procedure

```
